COMSATS University Islamabad,
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TScore

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Bachelor of Science in Computer Science (2021-2025)

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TScore

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By

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Abdul Hanan | **M. Adeel Sarwar**

CERTIFICATE OF APPROVAL

This is to certify that the final year project of BS(CS) **“TScore”** was developed by Abdul Hanan **(CIIT/FA21-BCS-070/SWL)** and M. Adeel Sarwar **(CIIT/FA21-BCS-094/SWL)** under the supervision of “**Dr. Muhammad Shoaib**” and and co supervisor “**Mr. Muhammad Jamil**” and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Computer Sciences.

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Executive Summary

In today's fast-paced digital world, people are constantly exposed to videos, news clips, and social media content that may or may not be true. With the rise of misinformation and fake news, it has become harder to identify which videos provide real facts and which ones are misleading. TScore is an AI-powered platform developed to assist in assessing the accuracy and trustworthiness of video content.

TScore is built to help users check the authenticity of video information by assigning a truth score, a clear, reliable number that indicates how accurate or factual a video is. The platform uses advanced technologies like Natural Language Processing **(NLP)**, Machine Learning **(ML)**, and content analysis to understand the spoken words in a video, translate them into the top 10 global languages, and verify the statements against real data sources. Users can easily sign up, log in, and upload videos they want to verify. Once uploaded, the system generates a transcript, and runs it through fact-checking algorithms. The result is a truth score shown on the dashboard. TScore also comes with several helpful modules.

A Translation Module for supporting multi-language users. User Features like commenting, liking, sharing, and reporting inappropriate content. Admin Tools for reviewing flagged videos, managing user accounts, and monitoring platform analytics. A Notification Service that keeps users updated about status changes, score results, or reports. A Feedback Module to help the system learn and improve based on user experience.

For transparency and safety, TScore also includes a report generation feature, allowing users to flag videos they believe are incorrect or harmful. Admins can then review these reports from their dedicated admin dashboard.

In short, TScore is more than just a video platform. It's a step toward building a more informed and responsible digital space. By combining advanced technologies like artificial intelligence, machine learning, natural language processing, and automated transcription, TScore helps users distinguish between real and misleading content. It encourages transparency by allowing community reporting, generates insights through dashboards and analytics, and supports global users with multilingual translations.

Acknowledgement

All praise is to the almighty Allah, who bestowed upon us a minute portion of his boundless knowledge by which we were able to accomplish this challenging task.

We are greatly indebted to our project supervisor **"Dr. Muhammad Shoaib".** Without their supervision, advice, and valuable guidance, the completion of this project would have been doubtful. We are deeply indebted to them for their encouragement and continual help during this work.

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Abbreviations

API	Application Programming Interface
CS	Computer Science
SRS	Software Require Specification
SDD	Software Design Description
GUI	Graphical User Interface
NLP	Natural Language Processing
IDEs	Integrated Development Environments

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Chapter 1

Introduction

1 Introduction

In today's digital age, content consumption has become a significant part of everyday life. However, the rise of misinformation poses a serious challenge to online platforms, leaving viewers uncertain about the credibility of the content they engage with. Content creators, while striving to produce valuable work, face the constant threat of having their content questioned, undermining their trustworthiness and audience reach. Traditional methods of verifying content's authenticity are often time-consuming and inefficient, leading to frustration among creators and viewers.

TScore is an innovative AI-powered platform designed to address these challenges and enhance the content-sharing experience. By combining advanced natural language processing (NLP) with machine learning, TScore acts as a digital lie detector for video content, verifying the accuracy of information and empowering users to make informed decisions. Content creators can upload videos with confidence, knowing that TScore will assess their content's authenticity, while viewers gain a clearer understanding of the trustworthiness of the media they consume.

With features like automatic content translation, ad monetization, and real-time performance analytics, TScore offers a comprehensive solution for creators to grow their reach while ensuring the integrity of their content. The platform's advanced AI capabilities enable seamless, cross-language content accessibility, allowing creators to reach a global audience without language barriers. Ad monetization provides creators with opportunities to earn revenue, ensuring that their hard work is rewarded while maintaining a seamless viewer experience. Real-time performance analytics offer valuable insights into viewer engagement, and interaction patterns, enabling creators to optimize their content strategy and tailor their approach to audience preferences.

1.1 Brief Overview

TScore is an AI-driven video streaming platform designed to verify the authenticity of video content, ensuring trust and transparency for both creators and viewers. Through its integration of AI and NLP, the platform analyzes video content for accuracy and rates it on a scale from 0–10. TScore also offers a range of features, including automatic content translation to reach a global audience, ad monetization options for creators, and detailed analytics to track viewer engagement. The platform promotes a community-driven environment by encouraging feedback and interaction, enhancing the overall experience for both creators and viewers. It empowers users to share ideas, collaborate on content, and support each other's growth.

1.2 Relevance to course modules

The development of TScore is closely linked to multiple courses from the Bachelor of Computer Science program, combining both theoretical knowledge and practical skills.

- **Artificial Intelligence and Machine Learning:** These courses form the foundation of TScore's truth-score generation system, enabling the platform to analyze and verify content authenticity using NLP and AI models.
- **Web Technologies:** Provided the technical expertise for building a responsive, user-friendly video streaming platform.
- **Human Computer Interaction and Software Design Methodologies:** Helped in designing intuitive user interfaces and planning system architecture to ensure a seamless user experience.
- **Database Systems and Software Engineering Concepts:** Guided the design of a robust backend system for handling user data, content, ads, and interaction logs efficiently and securely.
- **Programming Fundamentals, Object Oriented Programming, and Data Structures and Algorithms:** Played a central role in implementing core functionalities like video uploads, scoring algorithms, content processing, and ad logic.
- **Design and Analysis of Algorithms and Operating Systems:** Supported the optimization of algorithmic workflows and management of backend operations, ensuring scalability and responsiveness.
- **Software Project Management and Senior Design Project I & II:** Provided project planning, risk assessment, and team collaboration strategies essential for managing the development life cycle of TScore.
- **Report Writing Skills and Professional Practices for IT:** Contributed to the documentation, presentation, and ethical considerations involved in AI-powered systems.
- **Islamic Studies, Pakistan Studies, and Introduction to Business:** Encouraged socially responsible innovation while aligning with cultural values and market strategies relevant to user engagement and monetization.

Together, these courses enabled a holistic approach to building TScore, a platform that integrates modern AI techniques, ethical design, and practical software development.

1.3 Project Background

In today's digital age, video content dominates how people consume information, yet it also opens the door to widespread misinformation. With billions of users online, distinguishing fact from fiction has become increasingly challenging. Traditional platforms often lack effective mechanisms for verifying content, leading to confusion.

To address this growing concern, TScore was developed as an AI-powered video streaming platform that prioritizes content authenticity. Built with Django, it uses Natural Language Processing (NLP) to analyze video content, extract transcripts, and evaluate the credibility of information through a 0–10 (Truth Score). This empowers viewers to make informed decisions while consuming content.

TScore also introduces a unique monetization model where creators earn revenue through ad views, not by embedding ads within videos, but by displaying non-intrusive ads separately on the interface. Every view counts, making the earning process transparent and user-friendly. The platform aims to balance innovation, trust, and usability, offering a powerful solution for ethical content sharing in the modern digital landscape.

1.4 Literature Review

In today's digital ecosystem, video-sharing platforms have become dominant sources of information and entertainment. However, with the rise of such platforms, the spread of misinformation has also grown, prompting the need for content verification mechanisms. Several platforms, like YouTube and Facebook, offer content moderation, but their primary focus remains on copyright strikes or policy violations, rather than real-time misinformation detection.

Recent research in Artificial Intelligence and Natural Language Processing (NLP) highlights growing efforts to combat fake news and unverified claims using machine learning algorithms. Tools such as ClaimBuster, Factmata, and Full Fact use NLP and human-in-the-loop methods to validate information but are not integrated into mainstream video platforms. Moreover, ad monetization has largely remained independent of content credibility, often rewarding engagement without regard to truthfulness.

TScore aims to bridge this gap by integrating truth score generation, ad-based monetization, and real-time analytics into a single platform. While platforms like TikTok and Instagram prioritize short-form content virality, TScore emphasizes content integrity and transparency, offering creators and audiences a more responsible video-sharing ecosystem.

1.5 Analysis from Literature Review

The reviewed literature and existing platforms reveal a critical shortfall in content credibility verification at scale. While AI-powered misinformation detection is a research priority, few consumer-facing products implement it in real-time, especially within a creator economy model. TScore distinguishes itself by merging content verification with viewer-based monetization, ensuring that creators are rewarded not just for views but for verified and meaningful content.

Additionally, while platforms like YouTube display ads within videos or before playback, TScore introduces a non-intrusive model where ads appear separately, maintaining viewer focus while tracking each view to ensure transparency in revenue. This addresses user frustrations around disruptive ad formats and enhances the ethical use of advertisements. TScore aligns closely with contemporary research trends but goes a step further by offering a unified solution to misinformation, monetization, and audience trust.

1.6 Methodology and Software Lifecycle

For the development of TScore, the Agile Software Development Methodology has been selected. It supports incremental, iterative, and collaborative development, perfect for a feature-rich platform that evolves with user feedback.

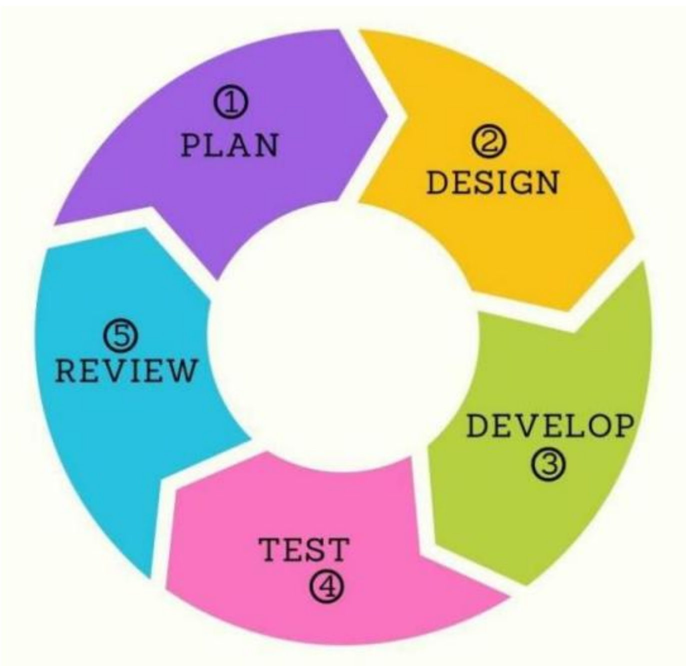

Figure 1.1: Agile Methodology Diagram

The Agile SDLC (Software Development Life Cycle) follows these stages:

- Requirement Gathering and Analysis
- Design and Prototyping
- Development in Iterations (Sprints)
- Testing and Quality Assurance
- Deployment
- Maintenance and Continuous Updates

This methodology allows for flexible planning, regular progress assessments, and the ability to incorporate changes without restarting the entire process and ideal for a project like TScore that may need to adapt to user preferences and emerging trends.

1.6.1 Rationale Behind Selected Methodology

The decision to use Object-Oriented Methodology in combination with the Agile lifecycle was driven by the complexity and modular nature of TScore. The system comprises several independent but interconnected components such as User Management, Video Uploading, Ad Display, Truth Score Analysis, and Performance Analytics.

Using this methodology enables clean architecture, better code management, and supports the scalable growth of the platform. It complements Agile's iterative nature, allowing different teams to work on different modules concurrently while maintaining consistency and integration across the system.

Chapter 2

Problem Definition

2 Problem Definition

2.1 Problem Statement

In the current digital age, video content has become a dominant mode of communication and information sharing. Most video-sharing services focus on popularity metrics such as likes, shares, and views, without evaluating the credibility or truthfulness of the content being shared. This leads to an environment where viral content is rewarded, even if it spreads false narratives, biased opinions, or incomplete facts. The absence of content verification mechanisms contributes to the rapid dissemination of misinformation, potentially influencing public opinion, social behavior, and even policy decisions.

Additionally, most platforms offer monetization opportunities that are tied directly to viewer engagement, regardless of the quality or accuracy of the content. Creators earn revenue even from misleading or harmful content, which encourages quantity over credibility. Another issue is the placement of advertisements within or before videos, which disrupts the viewing experience and adds no real control to the viewer.

There is no existing system that effectively verifies the factual integrity of video content while also providing non-intrusive ad-based earning opportunities. This creates a gap for a platform that can combine content validation, viewer-driven earnings, and ethical ad placement thus restoring trust in online content consumption and rewarding creators based on value and credibility. [1]

2.2 Deliverable and Development Requirements

2.2.1 Deliverables

- A web-based platform called TScore for uploading and viewing videos.
- A Truth Score Generation Module using NLP and AI techniques to evaluate video content.
- User Profiles with roles such as viewer, uploader, and admin.
- An Ad Display Module that shows non-intrusive ads (not within the videos).
- A Monetization System where earnings are generated through ad views.
- A Dashboard for analytics and performance tracking.
- A Secure Authentication System for all users.
- Admin Control Panel for managing users, content, and ad campaigns.

2.2.2 Development Requirements

To develop the TScore platform, a robust set of tools and technologies has been selected to ensure performance, scalability, and efficient development across all layers of the system:

- **Frontend:** The user interface is built using HTML, CSS, and JavaScript to ensure responsive design, cross-browser compatibility, and an engaging user experience.
- **Backend:** The server-side logic and API endpoints are handled using Python with the Django web framework, which offers rapid development capabilities and built-in security features.
- **Database:** PostgresSQL is used as the relational database to store and manage structured data efficiently, including user profiles, video content metadata, verification logs, and ad statistics.
- **Version Control & Collaboration:** Git and GitHub are employed to manage source code, track changes, and collaborate effectively throughout the development lifecycle.
- **AI & Language Processing Tools:** Integration with Gemini API and LangChain enables advanced natural language processing (NLP) tasks, such as content analysis and truth score generation.

This combination of technologies allows TScore to maintain high availability, modular design, and adaptability to future enhancements, including advanced ML-based verification systems and broader third-party API integrations.

2.3 Current System

Most existing platforms like YouTube, TikTok, or Facebook Watch focus primarily on entertainment, virality, and monetization through intrusive advertisements embedded within or before video playback. They offer no built-in mechanism for assessing the truthfulness of the video content or rewarding creators based on content accuracy or quality. This creates an ecosystem where engagement is prioritized over integrity, and misinformation spreads easily, especially through short-form viral content.

TScore addresses this critical gap by integrating AI-powered truth assessment mechanisms that analyze video transcriptions, evaluate factual accuracy using trusted sources, and generate a transparent “Truth Score.” This score empowers viewers to make more informed decisions about the content they consume, while encouraging creators to prioritize credibility. Unlike traditional platforms, TScore rewards creators not just for reach but for responsibility, promoting a healthier digital information space.

Moreover, by combining multilingual accessibility, community feedback, and ethical ad placement, TScore redefines the relationship between creators, viewers, and platform algorithms. It shifts the paradigm from algorithmic sensationalism to value-driven content delivery, building an ecosystem where factual integrity, user trust, and digital responsibility form the core of video-sharing culture.

Chapter 3

Requirement Analysis

3 Requirement Analysis

The requirement analysis for TScore identifies the essential functionalities needed to deliver a seamless and purposeful experience to different user roles including Admin and User. The system must support secure authentication, content creation and management, truth score evaluation, transcript services, advertisement viewing for earnings, and interactive features like likes, comments, and analytics. These requirements ensure the platform remains user-focused, functional, and transparent in its content credibility approach.

3.1 Use Case

A use case diagram is a simple yet powerful way to understand how users interact with a system. It visually represents the main features a system offers and the roles, called “actors”, that interact with those features. This kind of diagram is helpful in the early stages of system design, as it sets the foundation for how the final product should function from the user’s point of view.

Figure 3.1 shows the use case diagram of the TScore system, where two key users are involved: the User and the Admin. The diagram highlights what actions each of these users can perform within the system.

On the left side, we see the User, who is the general content creator or viewer. They have access to features like Authentication (signing up or logging in), Content Creation (uploading videos), Truth Score Analysis (getting a reliability score on their video), Transcript Translation, Reporting Content, and Sending Feedback. These actions reflect the primary goals of users, participating in the platform, contributing content, and ensuring a transparent environment.

On the right side is the Admin, who has more control over the system. The Admin can Manage Content (approve or remove videos), Oversee the Truth Score Analysis, Handle Reports, Translate Transcripts, and most importantly, Manage Users. The admin plays a key role in maintaining the platform’s integrity and ensuring that misleading or harmful content is properly dealt with.

This diagram gives a top-level view of the system's functionality. It helps both developers and stakeholders understand who does what and how different parts of the system are expected to interact. It also ensures that important functionalities aren’t overlooked during development.

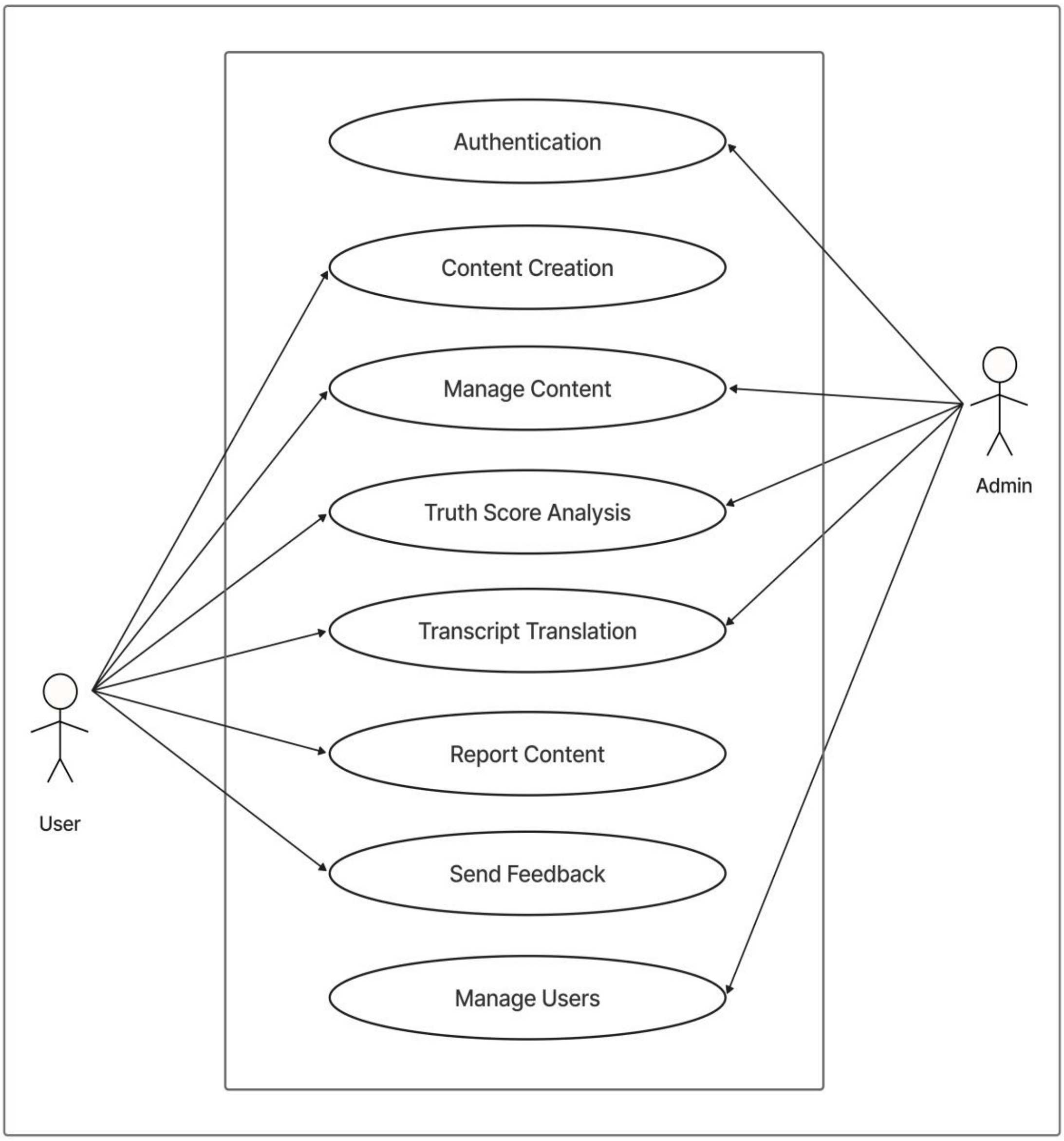

Figure 3.1: Use Case Diagram

3.2 Use Case Descriptions

The table below indicates comprehensive use case descriptions.

Table 3.1: Use Case Authentication

Use Case ID	UC01
Use Case Name	Authentication
Actors	User, Admin
Description	Enables users and admins to register and log in securely to the platform.

Trigger	User/Admin navigates to the login or registration page.
Preconditions	Internet connection is available. User/Admin is on the platform.
Postconditions	User/Admin is successfully logged in or registered.
Normal Flow	• User/Admin opens login page. • Enters credentials. • System verifies details. • Access granted.
Alternative Flow	If credentials are invalid, an error message is shown and reattempt is allowed.

Table 3.2: Use Case Content Creation

Use Case ID	UC02
Use Case Name	Content Creation
Actors	User
Description	Allows users to upload video content to the platform.
Trigger	User clicks "Upload Video" button.
Preconditions	User must be authenticated and logged in.
Postconditions	Video content is uploaded and stored for processing.
Normal Flow	• User selects video file. • Adds title, description. • Clicks upload. • System stores video.
Alternative Flow	If video format is invalid or file size too large, upload is rejected with message.

Table 3.3: Use Case Manage Content

Use Case ID	UC03
Use Case Name	Manage Content
Actors	Admin

Description	Admin can view, approve, remove, or flag uploaded video content.
Trigger	Admin logs in and accesses content panel.
Preconditions	Admin must be authenticated.
Postconditions	Video content is either updated, removed, or flagged.
Normal Flow	• Admin views content list. • Selects a video. • Performs action (approve/delete/flag).
Alternative Flow	If content list fails to load, admin is notified and can retry.

Table 3.4: Use Case Truth Score Analysis

Use Case ID	UC04
Use Case Name	Truth Score Analysis
Actors	System (background process), User
Description	System analyzes transcript of video and assigns a truth score based on data comparison.
Trigger	Video is uploaded and processed.
Preconditions	Transcript must be generated and processed.
Postconditions	Video is labeled with a truth score.
Normal Flow	• System extracts transcript. • Applies ML/NLP models. • Compares with reliable data. • Assigns score.
Alternative Flow	If analysis fails, a message is logged for admin review.

Table 3.5: Use Case Transcript Translation

Use Case ID	UC05
Use Case Name	Transcript Translation
Actors	User
Description	User can translate video transcript into top 10 supported languages.

Trigger	User selects a language option from the transcript panel.
Preconditions	Transcript must be generated.
Postconditions	Transcript is displayed in selected language.
Normal Flow	• User opens transcript. • Selects language. • System returns translated version.
Alternative Flow	If translation service fails, error message is displayed.

Table 3.6: Use Case Make Reports

Use Case ID	UC06
Use Case Name	Make Reports
Actors	User
Description	User can report a video they believe is misleading, harmful, or inappropriate.
Trigger	User clicks "Report Video" on video page.
Preconditions	User must be logged in.
Postconditions	Report is submitted and added to admin review queue.
Normal Flow	• User selects report option. • Chooses reason and submits. • System logs report.
Alternative Flow	If report submission fails, system notifies user.

Table 3.7: Use Case Send Feedback

Use Case ID	UC07
Use Case Name	Send Feedback
Actors	User
Description	Users can submit feedback about the platform's performance or suggestions.
Trigger	User clicks on "Feedback" section.

Preconditions	User is logged in.
Postconditions	Feedback is stored and sent to admin panel.
Normal Flow	• User writes feedback. • Submits form. • System stores it for admin.
Alternative Flow	If feedback field is empty, form is not submitted.

Table 3.8: Use Case Manage Users

Use Case ID	UC08
Use Case Name	Manage Users
Actors	Admin
Description	Admin can view, block, or delete user accounts.
Trigger	Admin accesses user management panel.
Preconditions	Admin must be logged in.
Postconditions	User accounts are updated based on admin actions.
Normal Flow	• Admin views user list. • Selects user. • Chooses action (block/delete/view details).
Alternative Flow	If user data fails to load, admin is alerted.

3.3 Functional Requirements

- **User authentication & role management:** Users sign up or log in securely, and their roles (user, admin, or content creator) are identified. Each role has access to specific system features based on permissions.
- **Content upload:** Users can easily upload their videos to the platform. This content is then analyzed and made available for viewers, subject to platform guidelines.
- **TruthScore analysis and display:** Uploaded videos are automatically analyzed using NLP and machine learning models to generate a truth score, which helps indicate the credibility of the content.

- **Auto transcription and translation:** Video speech is automatically transcribed and translated into multiple languages to improve accessibility and reach a global audience.
- **Content management (edit/delete):** Users and admins can manage content by editing or removing videos. This helps keep the platform clean, relevant, and aligned with community standards.
- **Advertisement integration & viewer rewards:** Ads can be displayed with content, and active viewers may earn rewards for engagement, which boosts retention and platform revenue.
- **Engagement features (likes, comments):** Viewers can interact with content using features like likes, comments, and shares, encouraging community participation and creator feedback.
- **Content analytics dashboard for users/admins:** Users and admins have access to dashboards showing insights such as views, likes, engagement rate, TruthScore distribution, and content performance over time.

3.4 Non-Functional Requirements

- **Scalability:** The system should efficiently scale to support a rapidly increasing number of video uploads and concurrent users without compromising performance.
- **Security:** The platform must ensure secure authentication.
- **Performance:** The system should deliver quick load times for videos, seamless playback, and responsive UI elements. Backend processes like video scoring, transcript fetching, and search indexing must be optimized for minimal latency.
- **Reliability:** The system must consistently return dependable TruthScore results and transcription outputs. Machine learning models should be trained on diverse data to ensure robustness, and error handling should be implemented to maintain service availability.
- **User-Friendly UI:** The interface should be intuitive and clean, ensuring that users of all tech backgrounds can easily navigate, upload videos, view analytics, and report content.

Chapter 4

Design and Architecture

4 Design and Architecture

4.1 System Architecture

The architecture diagram gives a clear picture of how the TScore system works behind the scenes. It shows the step-by-step journey of a video from the moment a user uploads it, all the way to getting a truth score and a translated transcript.

This diagram helps us understand how different parts of the system connect and talk to each other. When a video is uploaded, it first goes to the backend server. From there, it is sent to special modules that generate transcripts, run truth score checks using intelligent models, and translate the content if needed. Once all that is done, the results are saved and shown to the user in a simple and interactive way. [2]

Figure 4.1 outlines the overall architecture of the TScore system, detailing the data flow from video input to the final truth score and translated transcript output. It visually communicates how user actions are handled by the backend logic, processed by intelligent modules, and presented back to the user via the UI in a seamless and meaningful way.

Further details about each module involved in this process are discussed in the next sections. Overall, this architecture is designed to be scalable, secure, and efficient. It allows future improvements to be easily added, like smarter scoring techniques or advanced analytics, without disrupting the current system.

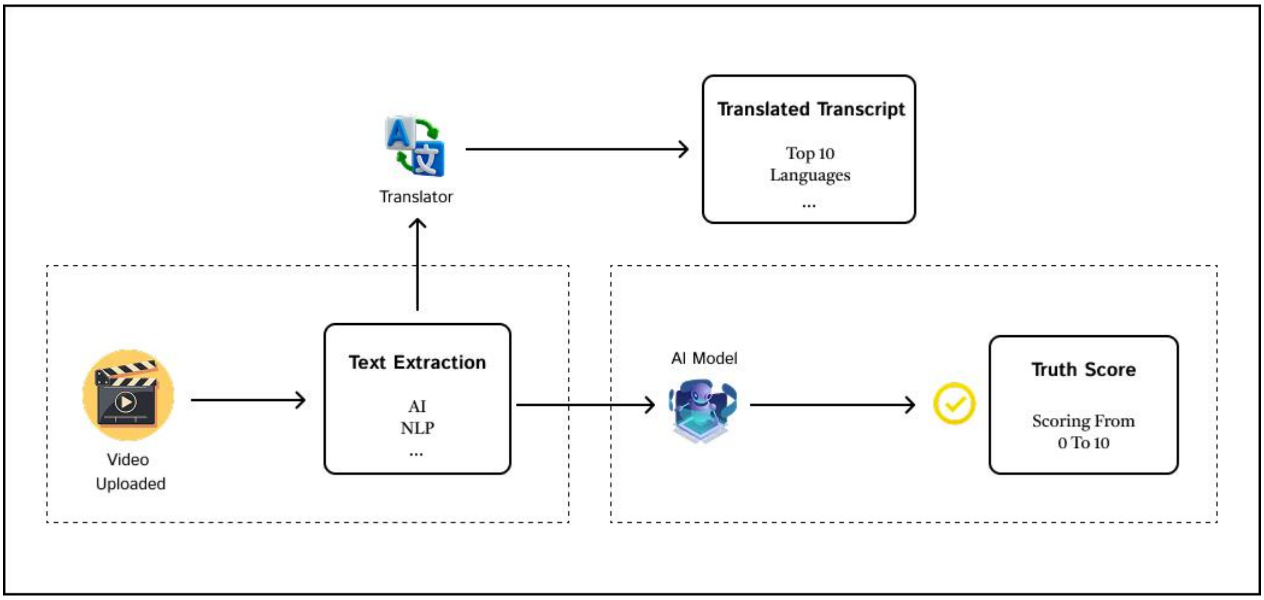

Figure 4.1: Architecture Diagram

Description: This diagram outlines the overall architecture of the TScore system, detailing the data flow from video input to the final truth score and translated transcript output.

Explanation:

- **Video Uploaded:** The process starts when a user uploads a video.
- **Text Extraction:** AI and NLP techniques are applied to extract speech or spoken text from the video.
- **Translator Module:** The extracted text is sent to a translator engine which translates it into multiple (top 10) languages for global accessibility.
- **Translated Transcript:** The result of the translation is displayed or stored as the translated transcript.
- **AI Model:** Simultaneously, the extracted text is processed by an AI model to analyze the authenticity and factual correctness.
- **Truth Score:** The AI model assigns a "Truth Score" to the video content, ranging from 0 (least truthful) to 10 (most truthful).

Purpose: This architecture highlights the modular nature of the system, where different components handle specific tasks while working together to assess and enhance the reliability and accessibility of video content.

4.2 Data Representation (Class Diagram)

The class diagram represents the core structure of the TScore system. It shows how different components of the system are organized and how they relate to each other. This diagram gives a blueprint of how data and functions are grouped in the backend, making it easier for developers to build, understand, and maintain the system.

In this diagram, each class represents a major element of the system like User, Video, Transcript, TruthScore, Feedback, or Admin. Inside each class, you can see its attributes and methods, that define what each component knows and what it can do.

This structure helps ensure that each feature in the TScore system has a clear and separate place, which avoids confusion and makes updates easier. It also supports clean coding practices, which is important as the project grows.

In the following sections, each of these components and their roles are explained in more detail to give a deeper understanding of how they contribute to the overall functionality of TScore.

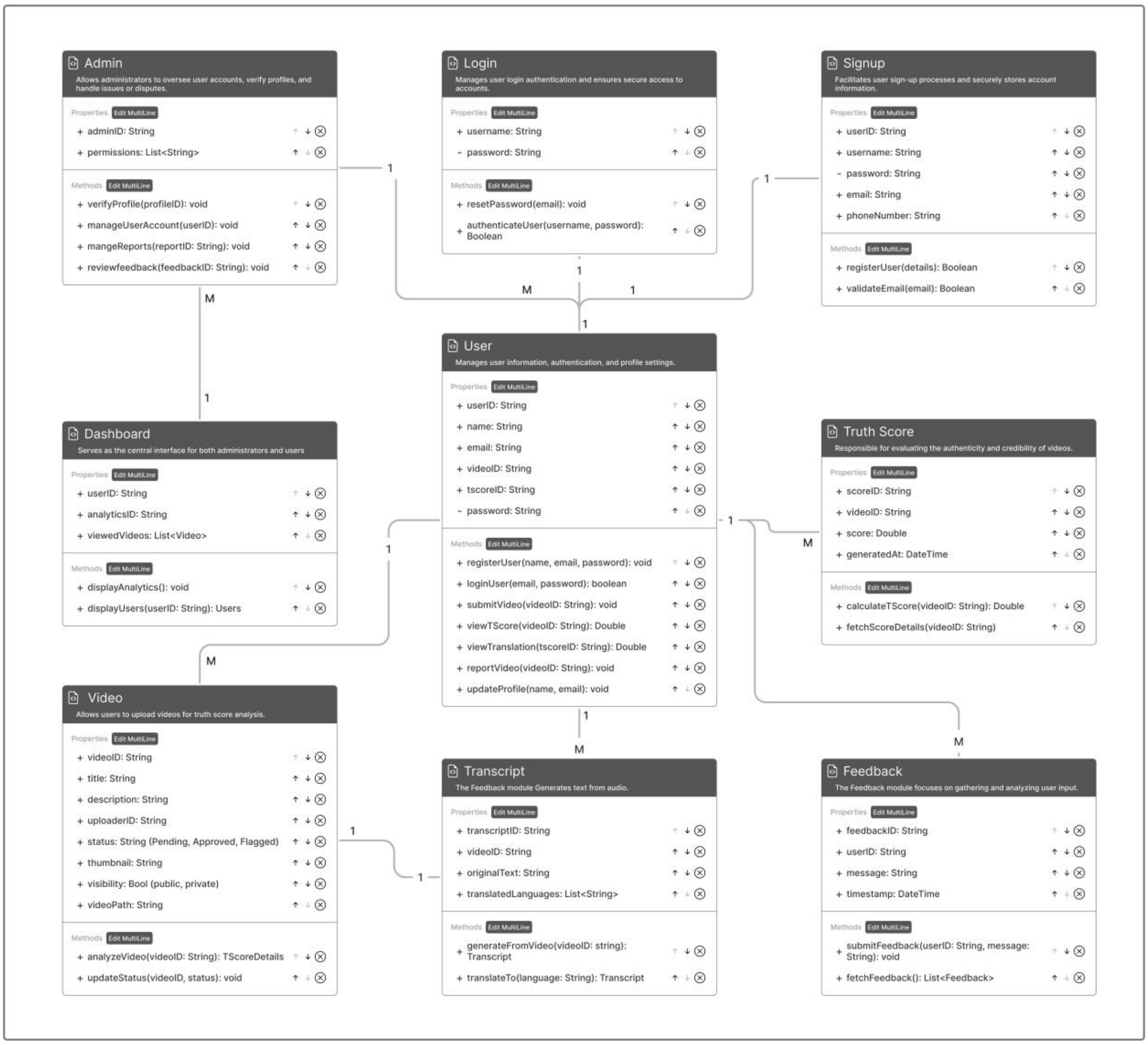

Figure 4.2: Class Diagram

Description: This diagram represents the structure of the software, detailing the different classes (e.g., User, Video, Transcript) and their attributes, methods, and relationships.

4.2.1 Components of the Diagram

The diagram represents the system's architecture through several interconnected modules. Each module has properties (data fields) and methods (functions), with relationships indicated by arrows and "1" or "M" to denote one-to-one or one-to-many associations. Below is a description of each component:

4.2.2 Admin Module

Purpose: Handles administrative tasks for managing users, videos, and feedback.

Properties

- userID: String (unique identifier for the admin)

- permissions: List<string> (list of permissions granted to the admin)</string>

Methods

- verifyProfile(userID: void (verifies user profiles)
- manageUserAccounts(userID: void (manages user accounts)
- manageReports(trending: String): void (handles reports, possibly related to trending content)
- reviewFeedback(feedbackID: String): void (reviews user feedback)

Relationships

- One-to-many (1:M) with User (admin manages multiple users)
- One-to-many (1:M) with Dashboard (admin accesses the dashboard)

4.2.3 Login Module

Purpose: Manages user authentication for secure access.

Properties

- username: String
- password: String

Methods

- resetPassword(email): void (resets user password via email)
- authenticateUser(username, password): Boolean (verifies login credentials)

Relationships

- One-to-one (1:1) with User (login is tied to a single user)

4.2.4 Signup Module

Purpose: Facilitates user registration and account creation.

Properties

- userID: String
- username: String
- password: String
- email: String
- phoneNumber: String

Methods

- registerUser(details): Boolean (registers a new user)
- validateEmail(email): Boolean (validates the email format)

Relationships

- One-to-one (1:1) with User (signup creates a single user)

4.2.5 User Module

Purpose: Stores and manages user information, authentication, and profile data.

Properties

- userID: String
- name: String
- email: String
- videoID: String
- tscoreID: String (links to truth score)
- password: String

Methods

- login(email, password): Boolean (user login)
- signup(videoID): void (associates a video during signup, if applicable)
- submitVideo(videoID): String (submits a video)
- viewTscore(videoID): String (views the truth score of a video)
- viewTranscript(videoID): String (views the transcript of a video)
- reportVideo(videoID): void (reports a video)
- updateProfile(profile): void (updates user profile)

Relationships

- One-to-many (1:M) with Video (user can upload multiple videos)
- One-to-many (1:M) with Truth Score (user can have multiple truth scores associated)
- One-to-many (1:M) with Feedback (user can provide multiple feedback entries)

4.2.6 Dashboard Module

Purpose: Serves as the central interface for both admins and users to view analytics and manage content.

Properties

- userID: String
- analyticsID: String
- viewVideos: List<video></video>
- displayUsers: List<user></user>

Methods

- displayAnalytics(userID): void (displays analytics for a user)

Relationships

- One-to-many (1:M) with Video (dashboard displays multiple videos)

4.2.7 Video Module

Purpose: Manages video uploads for truth score analysis.

Properties

- videoID: String
- title: String
- description: String
- uploadID: String
- status: String (Pending, Approved, Flagged)
- thumbnail: String
- visibility: Bool (public, private)
- videoPath: String

Methods

- analyzeVideo(videoID): String (analyzes the video content)
- updateStatus(videoID, status): void (updates the video status)

Relationships

- One-to-many (1:M) with Transcript (a video can have multiple transcripts)
- One-to-many (1:M) with Truth Score (a video can have multiple truth scores)

4.2.8 Truth Score Module

Purpose: Responsible for evaluating the authenticity and credibility of videos.

Properties

- tscoreID: String
- videoID: String
- score: Double (credibility score)
- generatedAt: DateTime

Methods

- calculateScore(videoID): Double (calculates the truth score)
- fetchScoreDetails(videoID): String (fetches details of the truth score)

Relationships

- Many-to-one (M:1) with Video (multiple truth scores can be linked to one video)

4.2.9 Transcript Module

Purpose: Generates text transcripts from video audio.

Properties

- transcriptID: String
- videoID: String

- originalText: String

Methods

- generate(from videoID): String (generates transcript from a video)
- translateTo(language): String (translates the transcript to another language)

Relationships

- Many-to-one (M:1) with Video (multiple transcripts can be linked to one video)

4.2.10 Feedback Module

Purpose: Collects and manages user feedback on the platform or videos.

Properties

- feedbackID: String
- userID: String
- message: String
- timestamp: DateTime

Methods

- submitFeedback(userID, message): void (submits feedback)
- fetchFeedback(): List<feedback> (retrieves feedback entries)</feedback>

Relationships

- Many-to-one (M:1) with User (multiple feedback entries can be linked to one user)

4.3 Detailed Description of the Diagram

Below is a detailed breakdown of how the system operates based on the diagram:

User Onboarding (Signup and Login)

- New users register through the Signup module, providing details like username, email, password, and phone number. The validateEmail() method ensures the email format is correct, and registerUser() creates a new user account.
- Existing users authenticate via the Login module using their username and password. The authenticateUser() method verifies credentials, and resetPassword() allows password recovery via email.
- Both modules interact with the User module, which stores user data like name, email, and associated videoIDs or tscoreIDs.

User and Admin Interaction

- The User module is the core of user interactions. Users can submit videos (submitVideo()), view truth scores (viewTscore()), access transcripts (viewTranscript()), report videos (reportVideo()), and update their profiles (updateProfile()).

- The Admin module oversees user management. Admins can verify profiles (verifyProfile()), manage accounts (manageUserAccounts()), handle reports (manageReports()), and review feedback (reviewFeedback()).
- The Dashboard module provides a unified interface for both users and admins to view analytics (displayAnalytics()) and manage videos/users (viewVideos, displayUsers).

Video Management and Truth Scoring

- Users upload videos via the Video module, which stores metadata like title, description, and status (Pending, Approved, Flagged). The analyzeVideo() method processes the video, and updateStatus() changes its status based on analysis or admin actions.
- The Truth Score module evaluates video credibility. It generates a score (calculateScore()) and stores it with a timestamp (generatedAt). Users can fetch score details using fetchScoreDetails().
- The Transcript module generates text from video audio (generate()) and supports translation (translateTo()), making the content accessible in multiple languages.

Feedback and System Improvement

- Users can provide feedback through the Feedback module, submitting messages (submitFeedback()) that are timestamped. Admins can retrieve and review feedback (fetchFeedback(), reviewFeedback()) to improve the platform.

4.3.1 Relationships and Data Flow

- The system uses a relational structure:
- A user can upload multiple videos (1:M with Video).
- A video can have multiple transcripts and truth scores (M:1 with Transcript and Truth Score).
- A user can submit multiple feedback entries (1:M with Feedback).
- Admins manage multiple users and access the dashboard (1:M with User and Dashboard).

This structure ensures scalability and efficient data management. It allows seamless integration of new modules as the system evolves. Additionally, it promotes maintainability by clearly separating concerns between components.

4.4 Process Flow/Representation

A flow diagram is a type of chart that visually represents the steps in a process or system using arrows and symbols. It shows how data or actions move from one stage to another and helps in understanding the overall workflow clearly and logically.

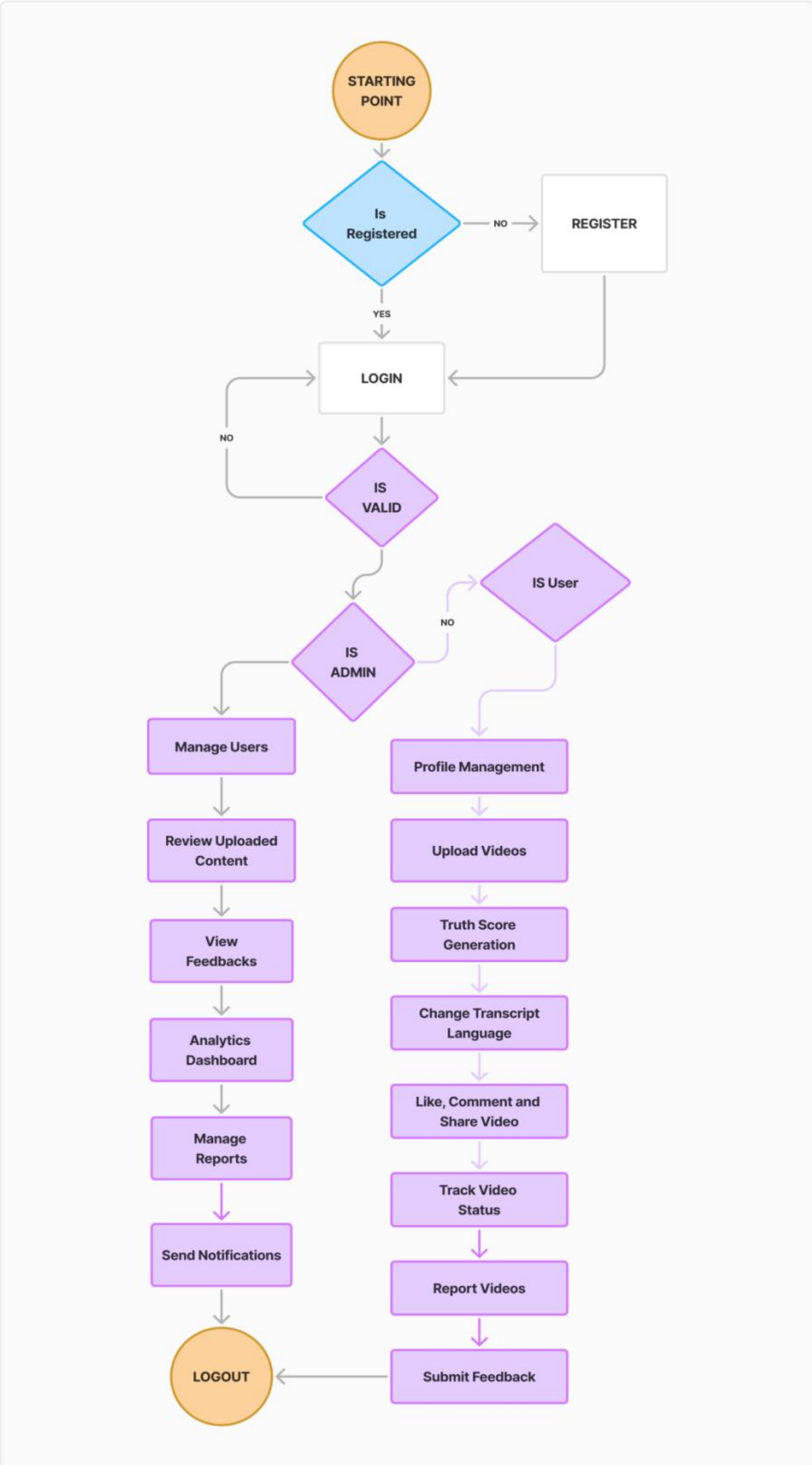

Figure 4.3: Flow Diagram

This flowchart effectively illustrates TScore's user-centric design philosophy and robust moderation framework. The visual representation clearly maps how various stakeholders, including content creators, viewers, and administrators, engage with the platform's ethical safeguards at each critical juncture. Beginning with user authentication, the flow progresses through content submission where videos automatically undergo AI-powered truth analysis, then branches to show multiple engagement pathways including reporting mechanisms for questionable content.

4.4.1 Key Processes

Authentication

Branches for new users (registration) and returning users (login).

Includes validation checks against database records.

Role-Based Access

Users

- Upload videos (triggers Truth Score generation).
- Interact via comments/likes.
- Report suspicious content.

Admins

- Review flagged content.
- Manage bans/notifications.
- Access analytics dashboards.

Core Features

- **Truth Score Generation:** Automated post-upload.
- **Feedback Loop:** User reports → Admin review → Content moderation.
- **Monetization:** Ad views tracked per video.

4.5 Design Models

The design model of TScore defines how different parts of the system interact to perform its core functions. It outlines the key roles of users and administrators, and how they engage with features such as video uploads, content analysis, truth score generation, and feedback submission. The use case representation highlights the various services offered to users and the responsibilities of the admin in managing the system. The sequence representation demonstrates how each process flows step by step. For example, how a video submitted by a user goes through text extraction, analysis, and scoring. Together, these models provide a clear picture of the system's behavior, interaction flow, and operational structure.

4.5.1 Sequence Diagram

A sequence diagram captures the flow of messages between different objects or components over time, making it ideal for understanding how a particular functionality or feature behaves step by step.

Figure 4.4 represents the sequence diagram of the TScore system, highlighting the communication between user, system, and backend components for actions like login, video upload, change language, and manage profile.

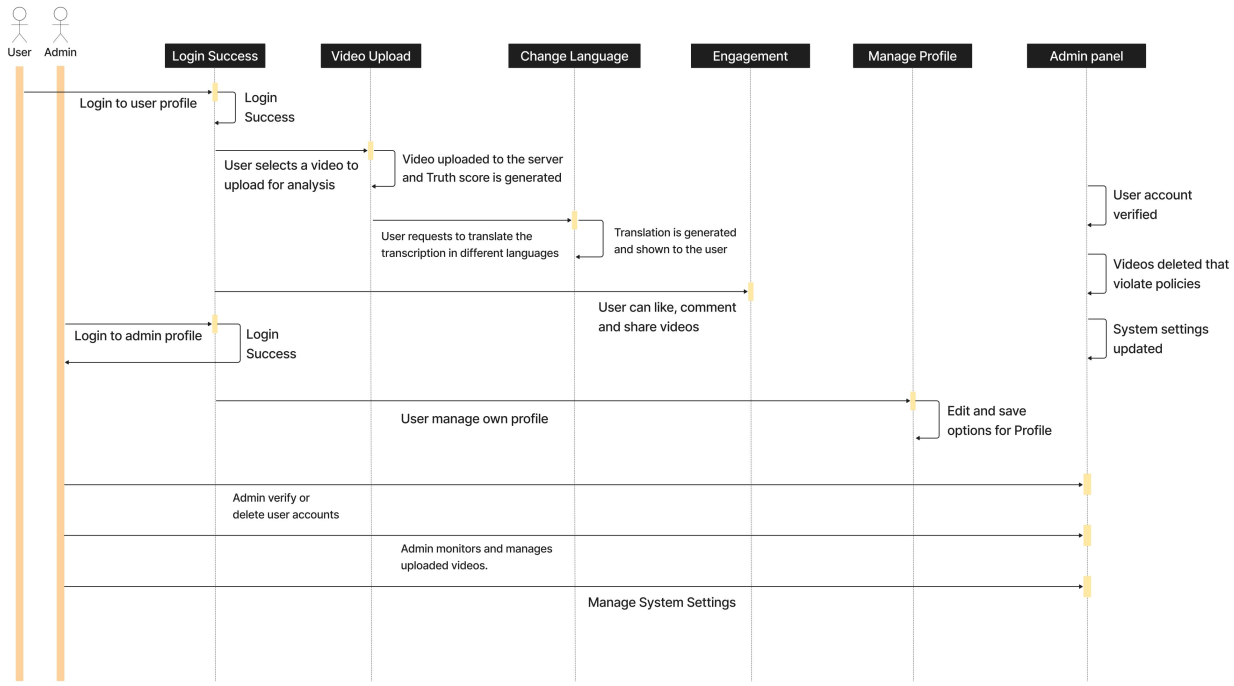

Figure 4.4: Sequence Diagram

The sequence diagram illustrates the interactions between two main actors (User and Admin) and the key modules of the TScore system during a typical workflow. It outlines the actions and communication flow across the following stages: user authentication, video upload and processing, truth score calculation, content moderation, and result presentation.

Initially, the User initiates the process by logging into the platform, where their credentials are verified by the authentication module. Upon successful login, the User uploads a video, which triggers the backend to receive and store the content securely. The video then undergoes processing through the Truth Score Engine, which utilizes advanced NLP and machine learning algorithms to analyze the video content and generate a truthfulness score.

Simultaneously, the Admin is notified of the new upload and can access the content moderation module to review flagged videos or those with low truth scores. This step ensures that any potentially misleading or false content is appropriately managed before being widely distributed. The system also maintains detailed logs of user actions and admin interventions for accountability and auditing purposes.

Finally, the processed results, including the truth score and any moderation status, are communicated back to the User and displayed on the video's interface, providing transparency and empowering users to make informed decisions about the content they consume. The sequence diagram effectively captures these dynamic interactions, highlighting the collaborative roles of Users, Admins, and system components in maintaining the platform's integrity.

Actors Involved

User: A regular user who can upload, view, and interact with video content.

Admin: The platform administrator is responsible for system maintenance and content moderation.

Modules Covered

- Login Success
- Video Upload
- Change Language (Translation)
- Engagement (Like, Comment, Share)
- Manage Profile
- Admin Panel

4.5.2 Step-by-Step Explanation

- Login Success
- User and Admin both initiate login actions.
- On successful login, they are redirected to their respective profiles.

Video Upload

- The user selects a video to upload for truth analysis.
- The video is uploaded to the server, and the Truth Score Generation Module processes it using AI and NLP.
- A truth score is generated and stored.

Change Language

- After uploading, the user can request the system to translate the video's transcription.
- The transcription is translated into the top 10 global languages.
- The translated content is then displayed to the user for wider accessibility.

Engagement

- Users can like, comment, and share videos.
- This promotes interactivity and community building around authentic content.

Manage Profile

- Users can edit and update their personal profile.
- All changes are saved to the system.

Admin Panel

- Verify user accounts.
- Delete videos that violate platform policies.
- Update system settings like algorithm parameters, UI configurations, etc.
- Monitor uploaded videos for moderation and truth scoring integrity.
- Delete or manage user accounts if necessary.

Chapter 5

Implementation

5 Implementation

In the implementation phase of TScore, we aim to deliver a reliable, intelligent, and user-centric platform for truth analysis of video content. This phase involves key steps to ensure secure operations, smooth user interaction, and efficient processing of AI-based truth scores.

Setup Development Environment

- Configure the development stack using Django (Python) for backend services, React for the frontend interface, and PostgreSQL for database management.
- Integrate NLP and AI libraries for processing and scoring video transcripts.
- Ensure responsive design and compatibility across desktops, tablets, and mobile devices for broader accessibility.

Database Design and Implementation

- Design and implement the database schema to manage users, videos, truth scores, comments, ad views, earnings, and roles.
- Create separate tables for users, video uploads, truth score results, comments, translations, and monetization logs.
- Optimize queries for fast data access and real-time dashboard updates.

User Authentication and Authorization

- Implement secure registration and login.
- Use Role-Based Access Control to manage different permissions for viewers, uploaders, and admins.
- Ensure session security, password hashing, and validation rules are applied across all forms.

User Interface Design

- Develop an intuitive, minimalistic UI using React with modular components for upload, analysis, profile management, and admin control. [3]
- Enable smooth navigation between core modules such as video upload, truth score viewing, translations, and user analytics.
- Integrate frontend with backend APIs for real-time content rendering and updates.

5.1 Modules

Modules are distinct components or phases of a project. Breaking down the project into modules helps in planning, organizing, and managing various aspects efficiently.

5.1.1 Content Creator Module

The Content Creator module empowers users to upload, manage, and monetize their video content on the TScore platform. Key functionalities include:

- **Signup & Profile Creation:** Allows content creators to register on the platform by submitting their personal and professional details.
- **Video Upload & Management:** Creators can upload videos directly to the platform, add titles, descriptions, tags, and thumbnail.
- **AI-Based Authenticity Scoring:** Every uploaded video is processed through an NLP pipeline, where AI analyzes speech content, compares it with verified sources, and generates a truth score (0–10 scale).
- **Monetization Management:** Enables creators to activate ad monetization on eligible content and view earnings analytics.
- **Insights & Analytics:** Provides real-time data on video performance, including views, watch time etc.
- **Earnings:** Users can earn money by watching ads.
- **Translation & Accessibility Tools:** Supports multilingual translations to increase content reach across diverse audiences globally.

5.1.2 Viewer Module

The Viewer module offers an intuitive interface for users to explore, watch, and evaluate video content while engaging with the community. Key functionalities include:

- **Account Creation:** User need to make a account for earning and engaege with videos.
- **Content Discovery:** Users can browse content based on categories, search by keywords, and follow their favorite creators.
- **Video Player with TScore Display:** Each video is embedded with a visible TScore truth rating and summary of AI findings, enabling informed viewing. User can also change transcript language.
- **Engagement Tools:** Viewers can like, comment, share videos, and provide feedback on AI scoring accuracy.
- **Earnings:** Users can earn money by watching ads.
- **Watch History & Personalization:** Tracks previously watched content and recommends new videos based on interests, trust scores, and interaction patterns.

5.1.3 Admin Module

The Admin module is the central management system that ensures smooth functioning and compliance of the platform. Key functionalities include:

- **User Management:** Allows administrators to manage creator and viewer profiles, verification, or removal based on policy violations.

- **Content Oversight:** Admins can review uploaded content flagged by users.
- **AI Performance Monitoring:** Monitors the accuracy and reliability of the AI-based truth detection algorithm.
- **Payment & Monetization Oversight:** Manages the monetization process, ad revenue distribution, and payment issues.
- **Community and Policy Management:** Handles moderation of user-generated content (e.g., comments), enforces community guidelines, and oversees platform-wide announcements.

5.1.4 AI & NLP Processing Module

This module is responsible for analyzing video content to determine the credibility and accuracy of informations. Key functionalities include:

- **Speech-to-Text Conversion:** Converts spoken video content into structured text using advanced speech recognition algorithms.
- **Prompt-Based Evaluation:** Applies predefined prompts and fact-checking logic to analyze extracted content for accuracy.
- **Truth Scoring Algorithm:** Generates a truthfulness score (0–10) based on cross-referencing with trusted sources, detecting misinformation patterns, and evaluating content tone.
- **Continuous Learning Engine:** Incorporates user feedback and expert reviews to refine the AI model, improve detection rates, and reduce bias.

5.1.5 Ad & Revenue Module

This module facilitates the monetization process. Key functionalities include:

- **Ad Integration:** Supports various ad types such as sponsored banners to generate revenue for users.
- **Payment Gateway Integration:** Connects with secure payment systems to facilitate revenue withdrawals.

5.2 Algorithm

The core algorithm of TScore revolves around evaluating the factual credibility of video content using AI. The system follows a structured process to extract and assess information from uploaded videos.

5.1.1 User Authentication & Input Collection

Users register and log into the system. After authentication, they upload a video along with essential metadata (title, description, tags).

5.1.2 Transcription Generation

Once a video is uploaded, the backend system extracts the spoken content and converts it into text using a speech-to-text engine.

5.1.3 Prompt Construction & Gemini Evaluation

The generated transcription is inserted into a predefined prompt and sent to the Gemini API, which performs fact-checking. The prompt instructs Gemini to:

- Analyze the text for factual accuracy.
- Cross-check claims using its knowledge base.
- Generate a truth score (0–10) reflecting the video's factual reliability.
- Provide supporting detail and reasoning behind the assigned score.

5.1.4 Storing Evaluation Results

The system stores the video, transcription, truth score, and evaluation details in the database. This data is linked with the user and becomes available for public view.

5.1.5 User Engagement Handling

Other users can view the video and engage through likes, comments, and follows. These interactions are stored and monitored for future analytics or moderation.

5.3 External APIs

Table 5.1: External APIs

Name of API	Description of API	Purpose of Usage	List of Functions
Gemini API (Google AI)	A powerful generative AI model capable of understanding and analyzing text and media.	Used to evaluate video transcript, description, and metadata to generate a reliable truth score.	evaluate_video(), generate_truth_score()
Google Maps API	Provides geolocation and mapping functionalities.	Used to show the origin or context of a video or channel's publishing location (if location-based).	fetch_location_details(), display_video_location()

5.4 User Interface

The user interface of TScore has been designed to prioritize usability, transparency, and an informative user experience. Throughout the development process, multiple key interface screens were created and refined to support both users and content creators. These screens not only reflect modern UI design principles but also ensure that navigating the platform is intuitive and consistent across all devices. [4]

One of the most important screens is the Home screen, which acts as the landing page for all users. It displays a curated grid of videos, each showing a preview thumbnail, title, views, upload time, and the creator's name. However, what makes TScore unique is the inclusion of a truth score badge at the top-left of each video. These truth scores are color-coded: green for reliable and factual content, red for false or misleading videos, and orange/yellow for moderately accurate content. This visual cue immediately informs the user about the trustworthiness of a video, encouraging responsible media consumption from the outset.

Another core screen is the Dashboard, which becomes accessible once a user logs in. This area allows users to manage their uploaded content, monitor engagement, and check the performance of their videos based on both views and truth scores. It offers creators a transparent view of how their content is being evaluated and provides an incentive to create meaningful and accurate videos.

To enhance user personalization, several supporting screens have been added. The Followings section allows users to follow trusted creators and easily access their content. The Saved Videos and Liked Videos sections let users manage their favorite or to-watch-later content. History enables users to view their past activity, ensuring ease of access to previously watched materials.

In addition to personal features, the platform includes public exploration sections like Trending, which displays the most talked-about videos on the platform, not solely based on views, but also on truth score performance. The Truth and False screens allow users to dive deep into content that has been verified for accuracy or flagged as misleading. These sections help build awareness and educate users on differentiating between factual and false narratives.

Furthermore, TScore also includes essential utility screens such as Settings, Report History, and Send Feedback. These areas empower users to control their experience, report content responsibly, and share suggestions for improvement. A clean and easily accessible Sign Up / Sign In screen has also been implemented to provide smooth onboarding for new users, while returning users can seamlessly continue their activity.

Each of these screens maintains a consistent sidebar layout, ensuring easy navigation throughout the platform. Icons and labels are clearly marked, and the user flow is kept simple to accommodate users of all technical skill levels. Visual hierarchy is maintained using spacing, contrast, and minimal color usage, keeping the interface clean and clutter-free. Following are some of the screens of TScore.

5.3.1 Userside Screens

The User SignUp screen allows new users to register by entering their name, email, password, and other required information. It includes input validation to ensure correct data and links to login for existing users.

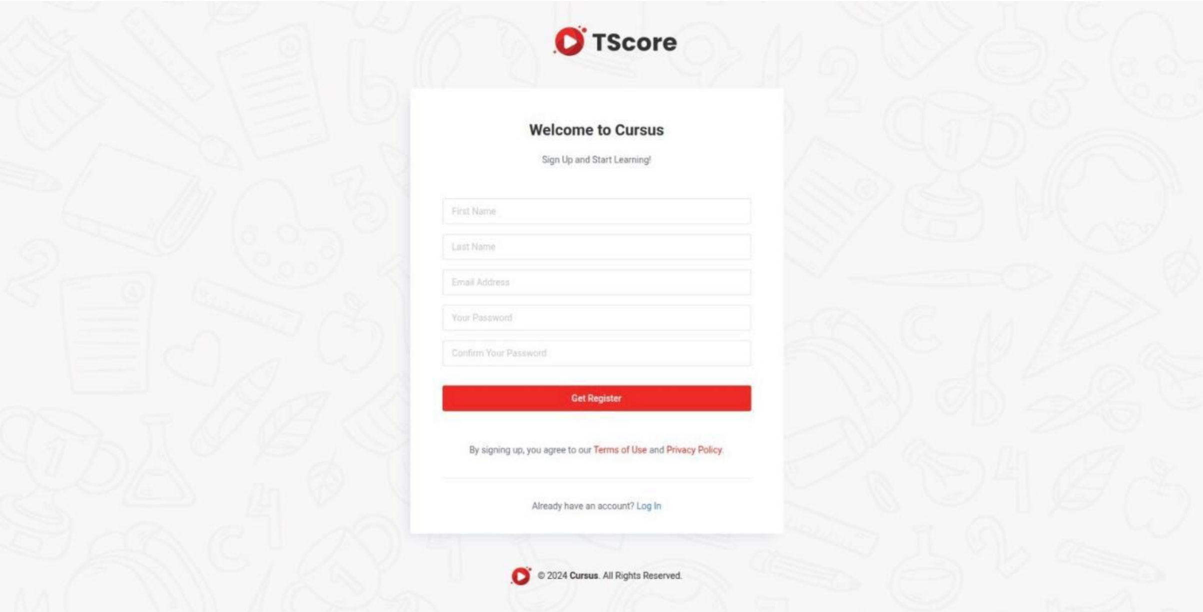

Figure 5.1: User SignUp

The User Login screen allows registered users to log into the TScore platform using their email and password. It includes a 'Sign Up' option and redirects to the signup screen if the user doesn’t have an account.

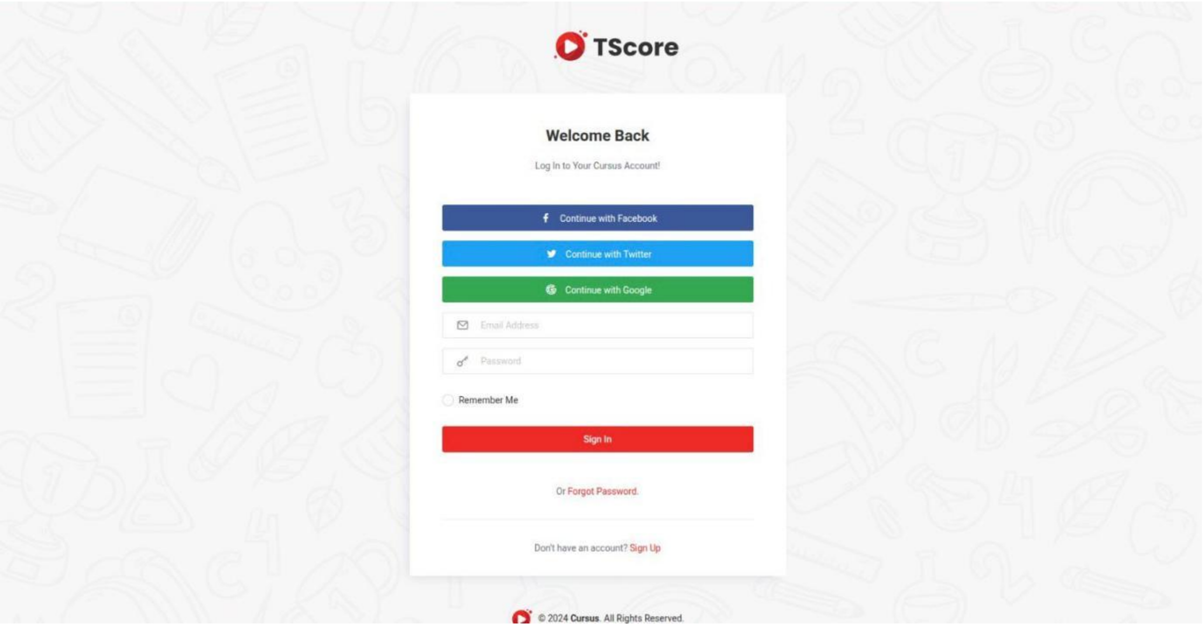

Figure 5.2: User Login

The Home Screen displays trending and recommended videos based on user preferences. It includes a search bar, navigation menu, and quick access to various video categories.

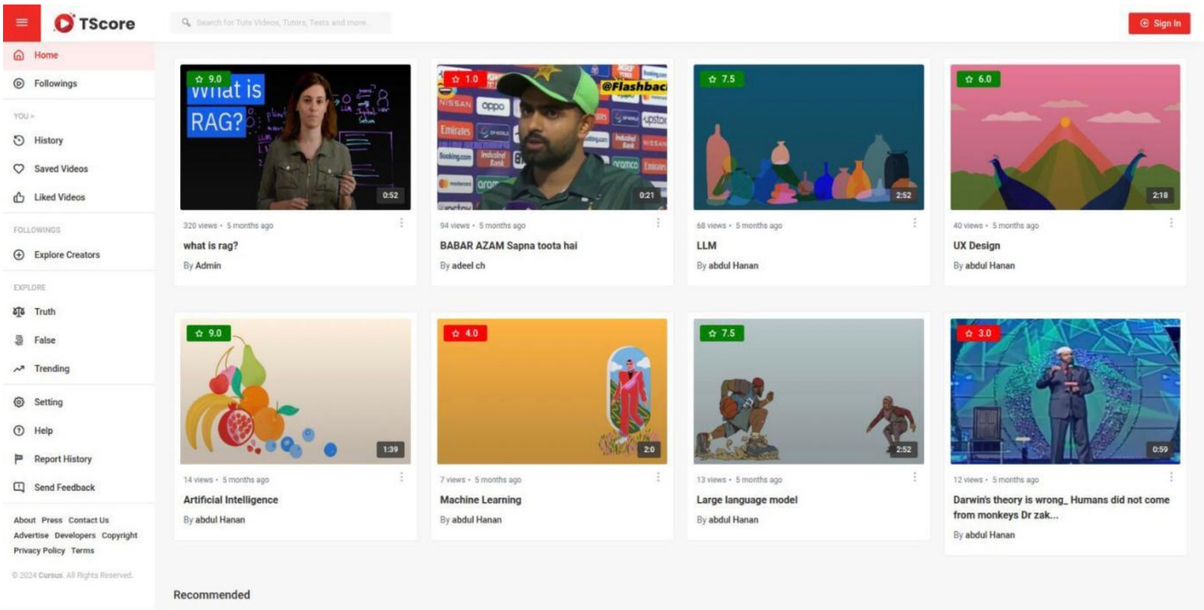

Figure 5.3: User Home Screen

Screen shown in Figure 5.4 provides information about the advertisements shown to users, including details of the advertiser, duration, and type of ad. It may also show engagement statistics.

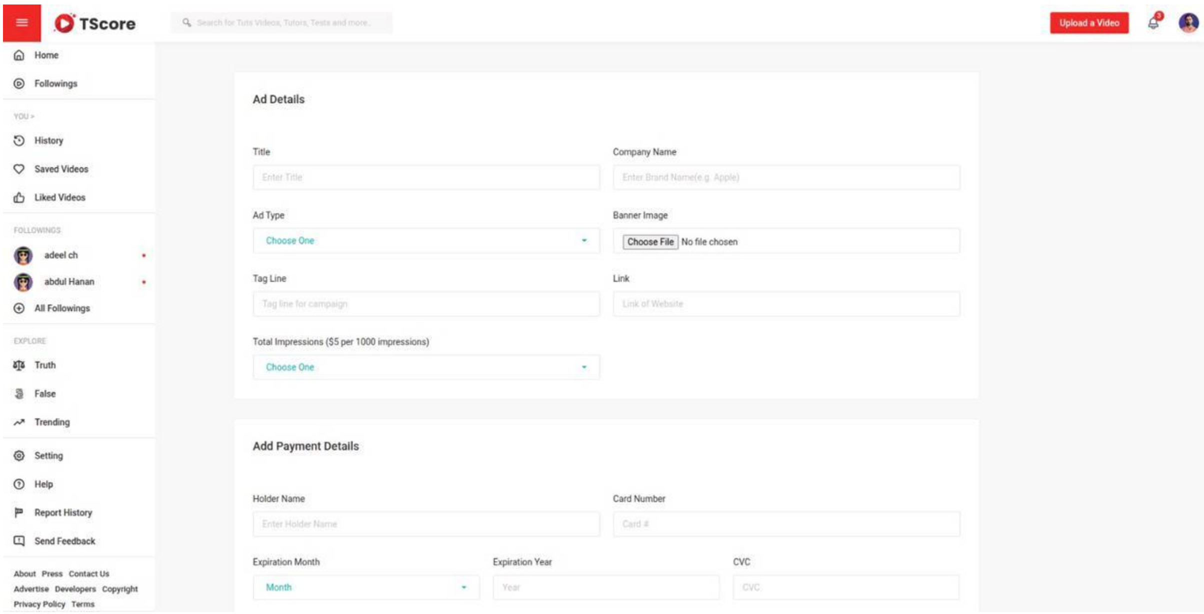

Figure 5.4: Ad details

The My Videos screen lists all videos uploaded by the user. It provides options to edit, delete, and view insights related to each video.

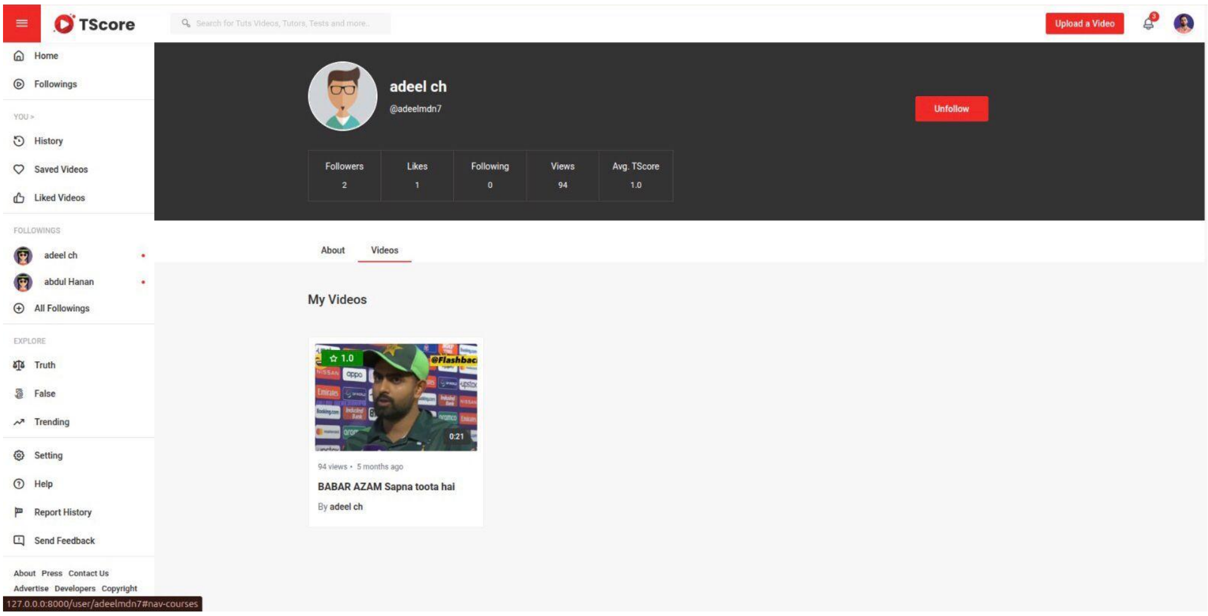

Figure 5.5: My Videos

Screen shown in Figure 5.6 shows a collection of all videos that the user has liked. Users can replay or remove videos from the liked list.

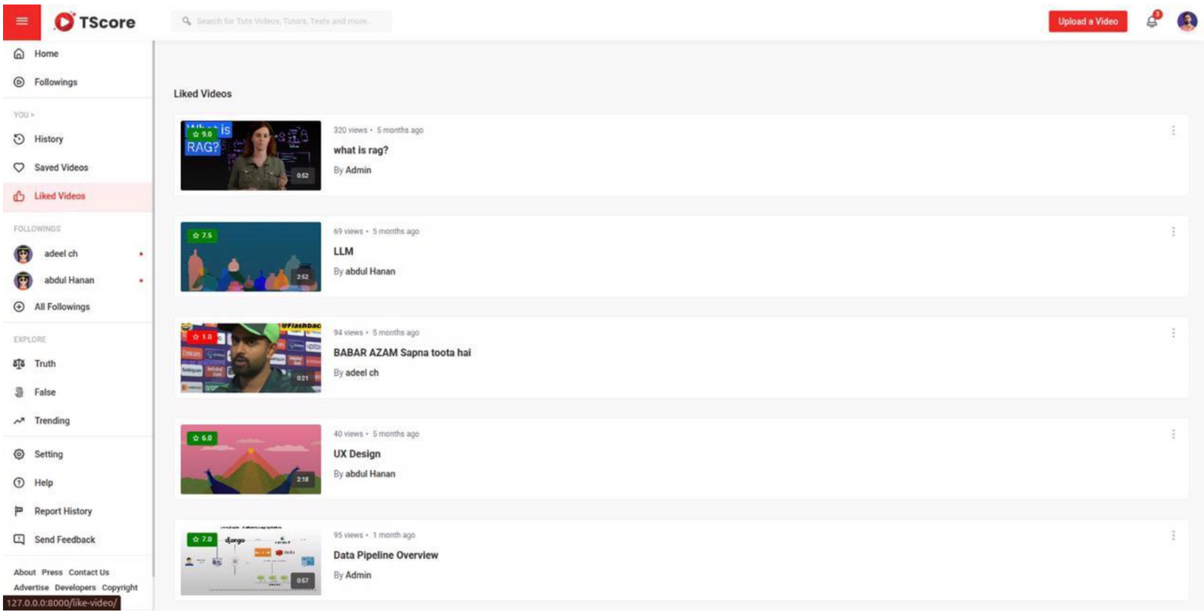

Figure 5.6: User Liked Videos

Watch History displays a list of videos the user has watched. Users can revisit their history from this screen.

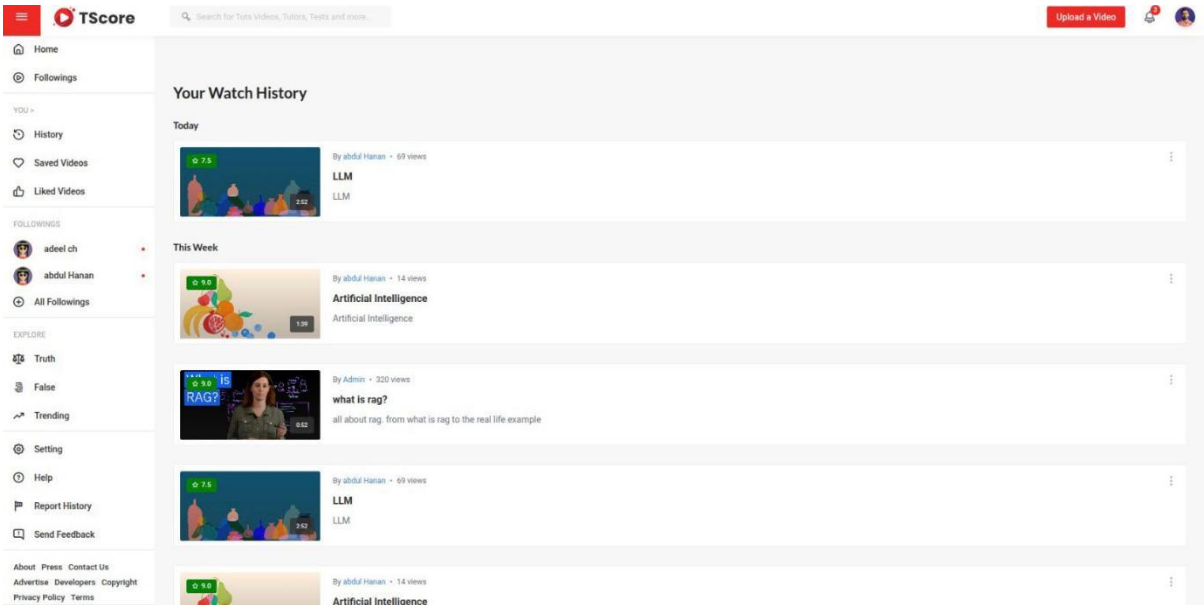

Figure 5.7: User Watch History

Saved Videos allows users to bookmark or save videos to watch later. These videos are accessible even if they are not currently trending.

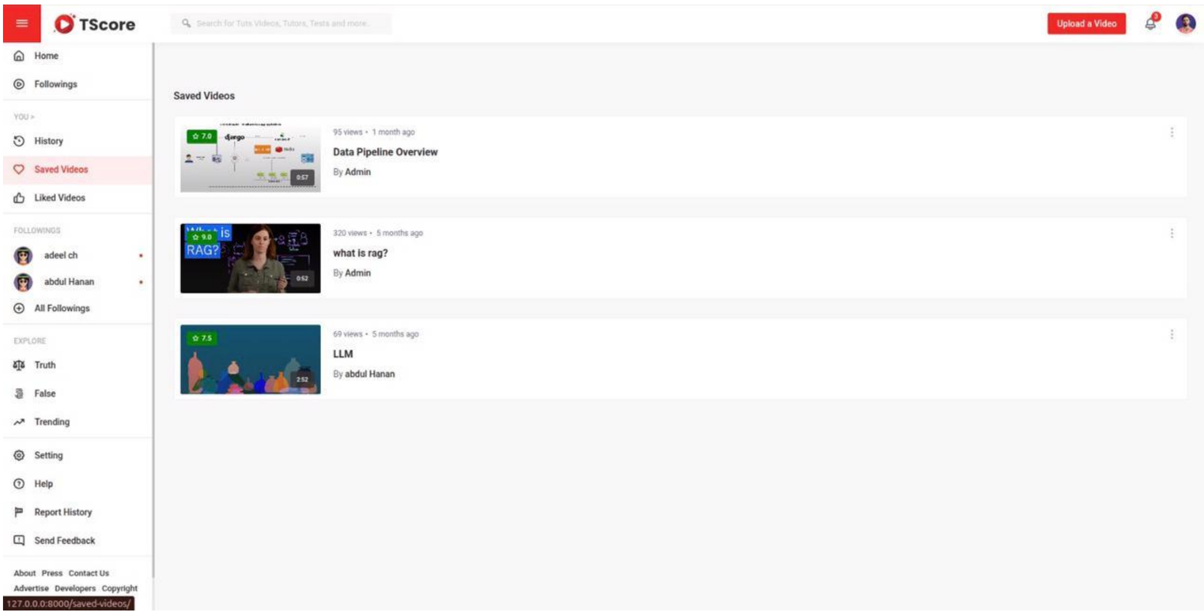

Figure 5.8: User Saved Videos

Screen shown in Figure 5.9 displays all the reports submitted by the user regarding misleading or inappropriate videos, along with their status and admin responses.

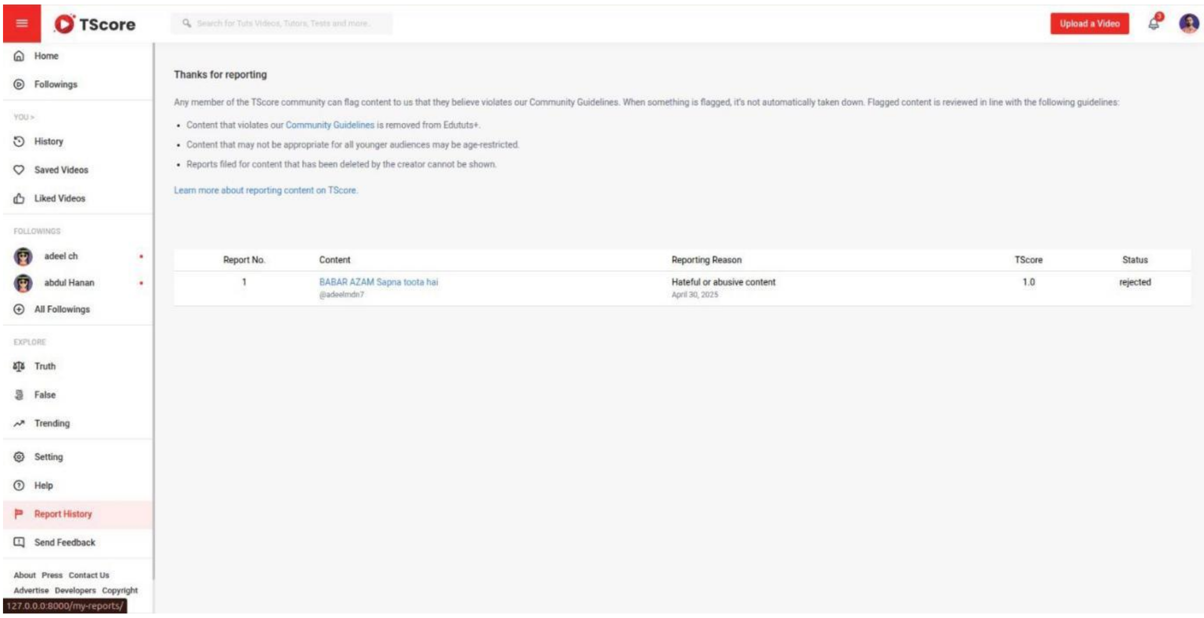

Figure 5.9: Report History of Videos

The False Video screen lists all videos with a truth score lower than 5 out of 10, indicating potentially misleading or false content.

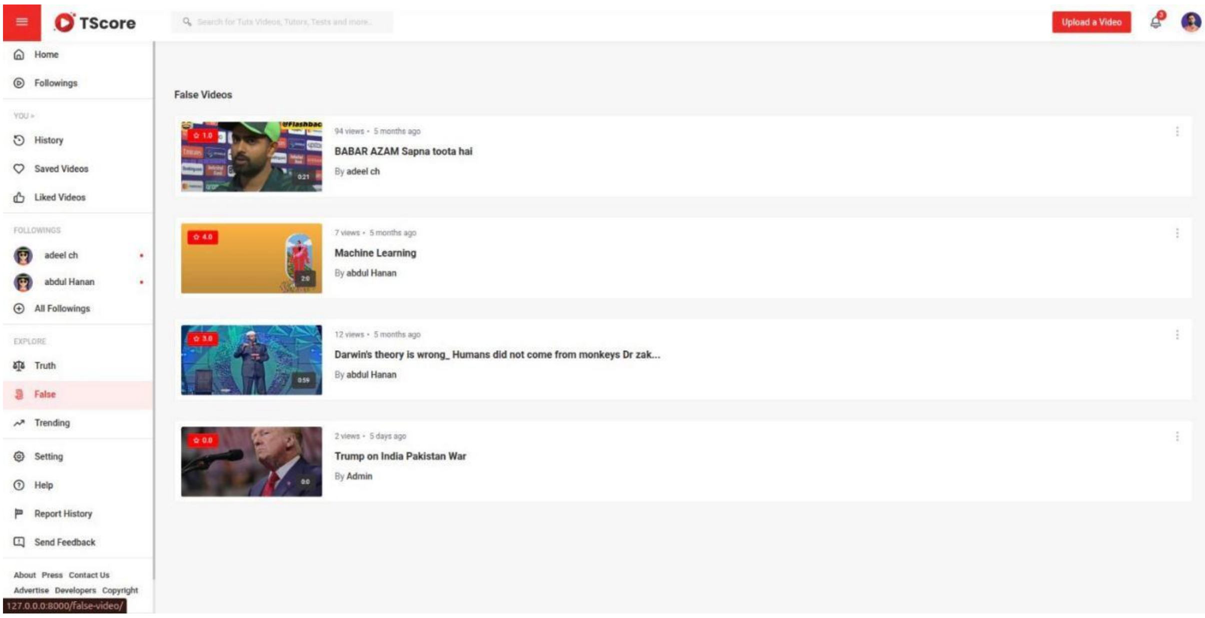

Figure 5.10: False Videos

Screen shown in Figure 5.11 shows videos with a high truth score (above 5), representing content considered reliable and factually accurate.

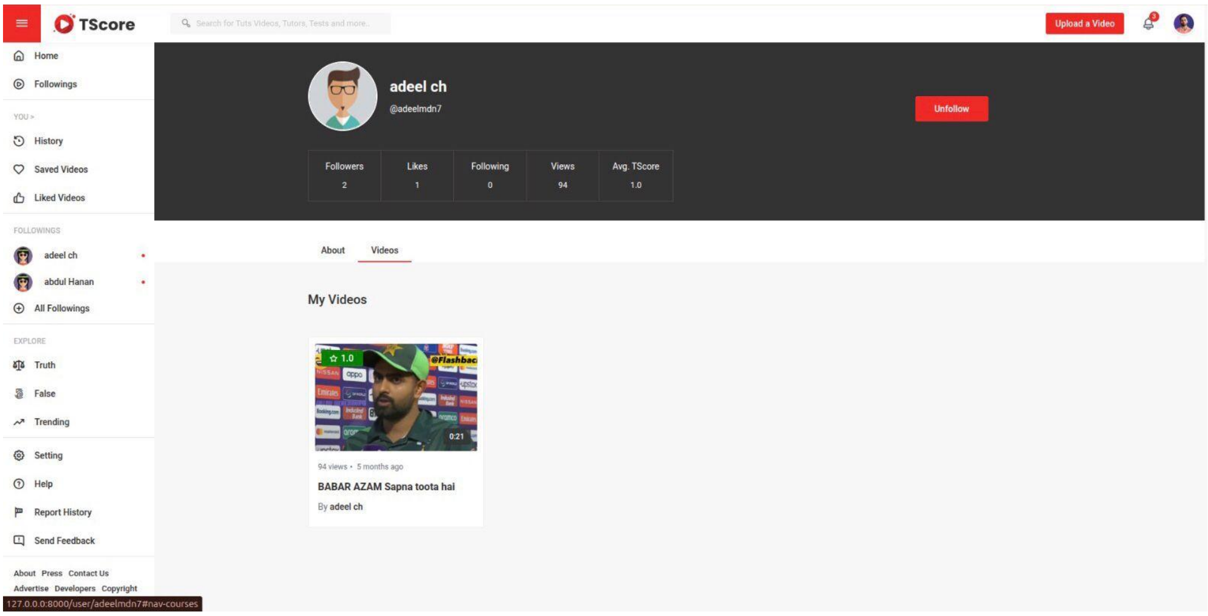

Figure 5.11: Truth Videos

Screen shown in Figure 5.12 displays the list of creators or users the current user is following, along with quick links to their content.

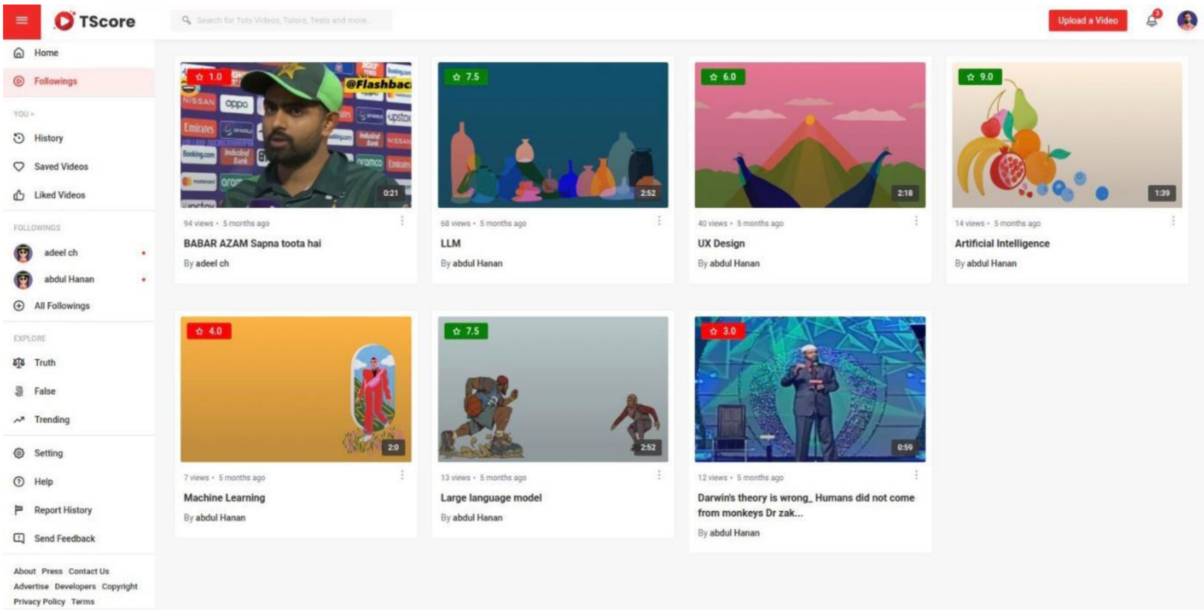

Figure 5.12: User Following Screen

Privacy Policies screen presents the platform's privacy policies regarding data collection, usage, and user rights.

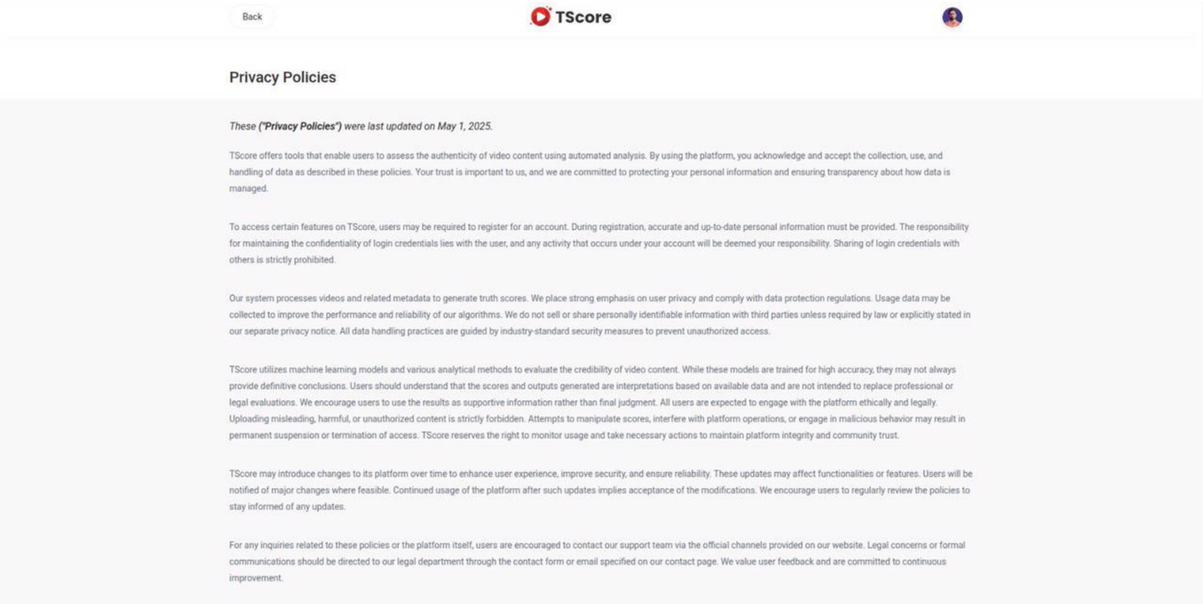

Figure 5.13: Privacy Policies

Screen shown in Figure 5.14 highlights the most popular and viral videos on the platform, ranked based on views, likes, and shares.

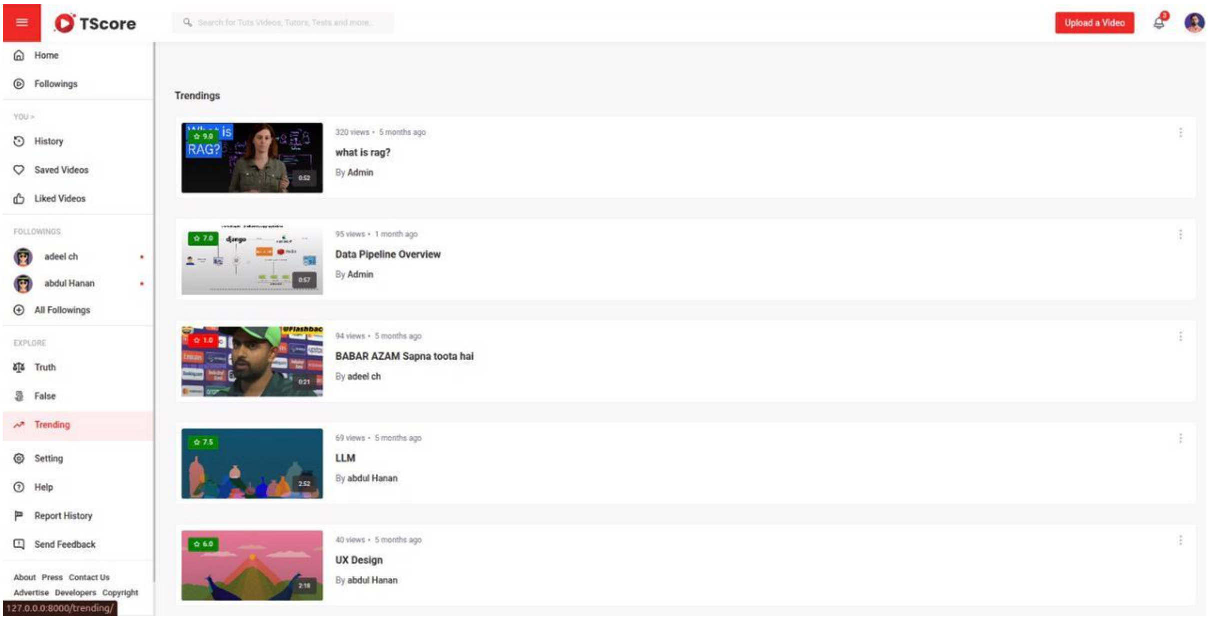

Figure 5.14: Trending Videos

The Help Center provides users with FAQs, guides, and support contact options to assist with common issues and questions.

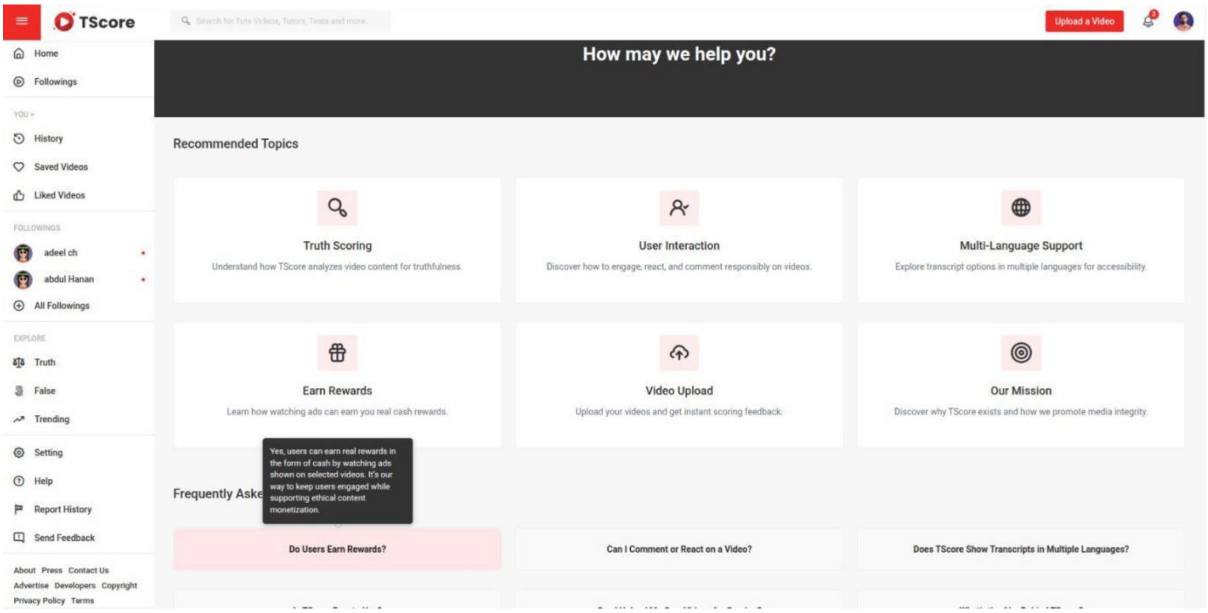

Figure 5.15: User Help Center

Screen shown in Figure 5.16 describes the core features of TScore, such as Real-time transcripts, truth scoring, translation, and content insights.

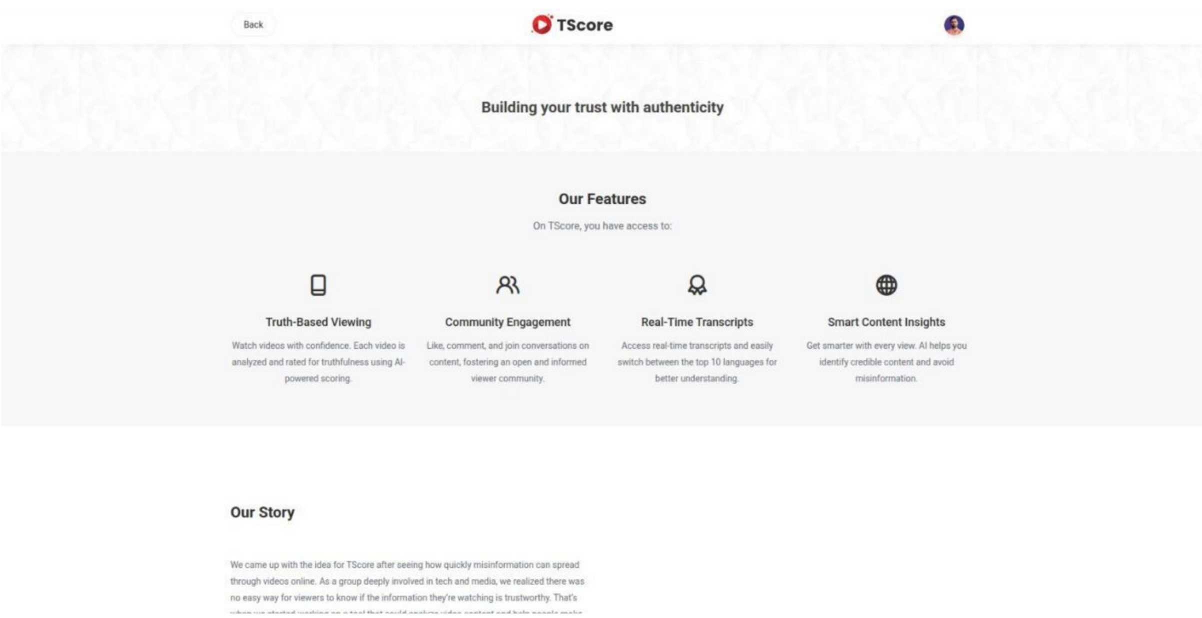

Figure 5.16: TScore Features Screen

Screen shown in Figure 5.17 allows users to send feedback, complaints, or suggestions directly to the TScore team for review and improvement.

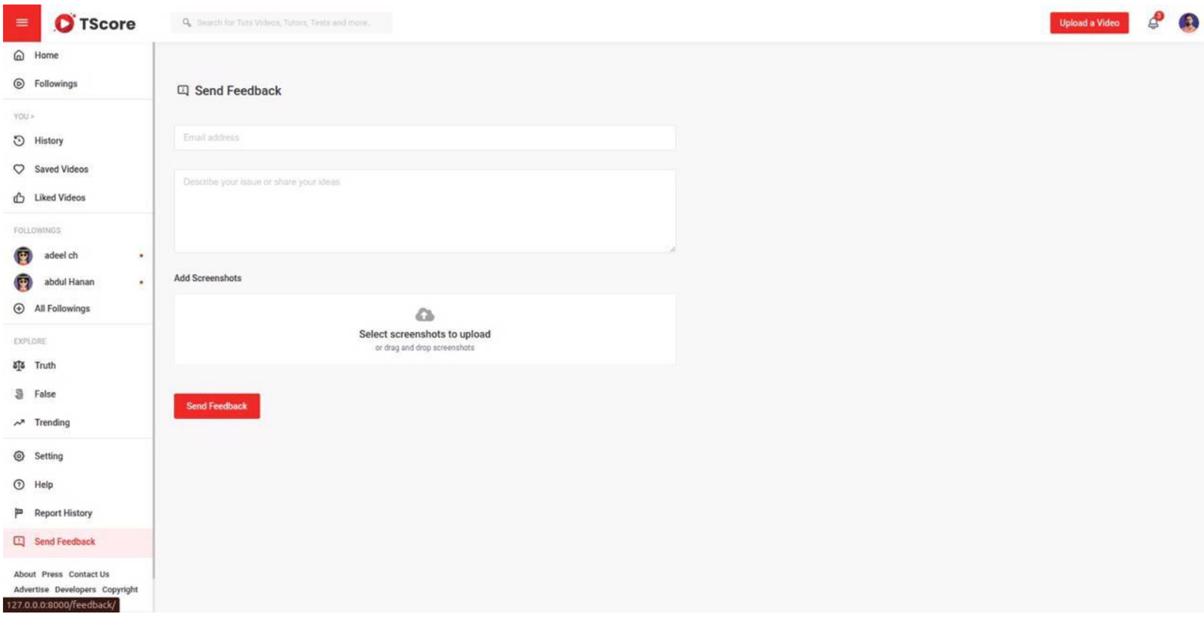

Figure 5.17: Send Feedback

Figure 5.18 displays TScore's contact details including email, phone number, and address.

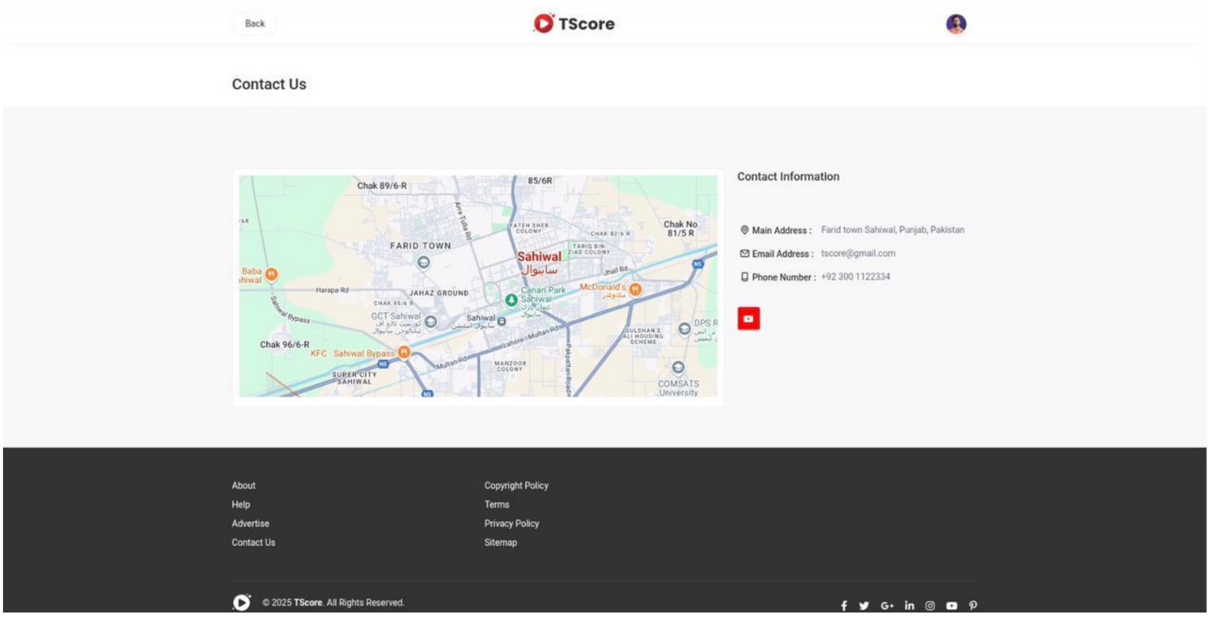

Figure 5.18: Contact Information

Figure 5.19 shows the platform's terms and conditions for using the service, outlining user responsibilities and prohibited behaviors.

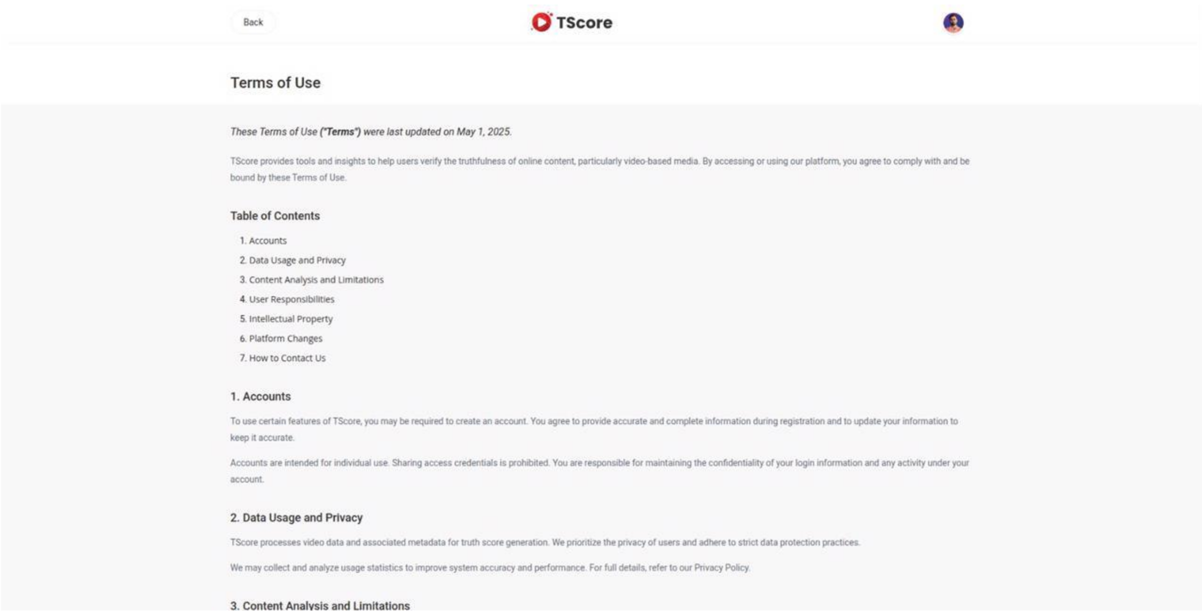

Figure 5.19: Term of Use

5.3.2 Admin Dashboard

Admin Dashboard provides a central interface for managing the platform including analytics, user management, and content oversight.

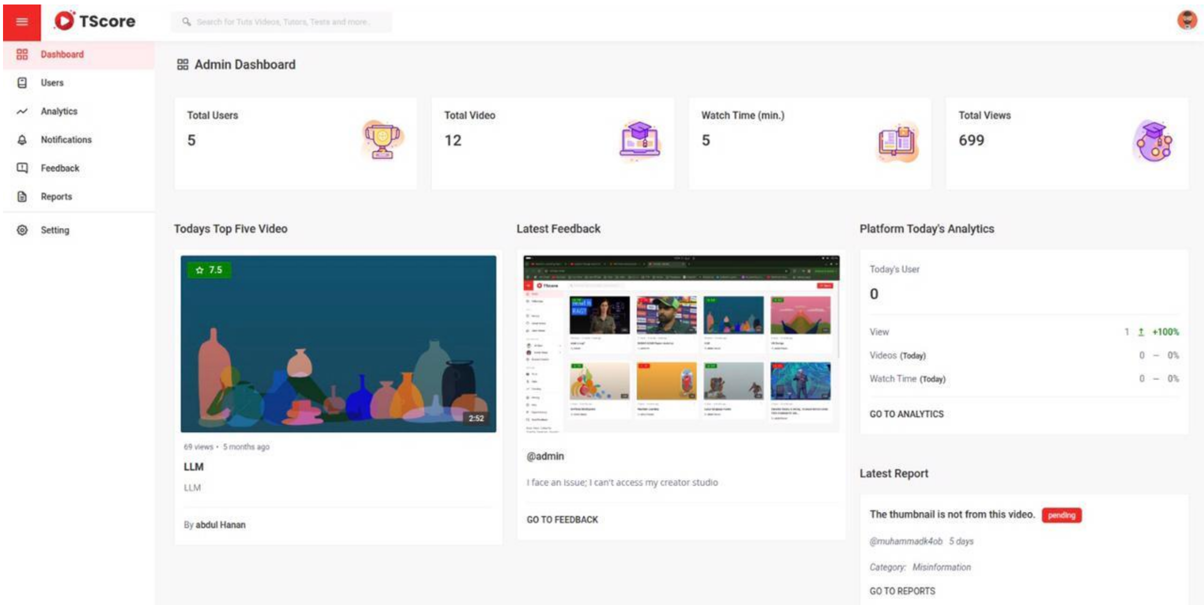

Figure 5.20: Admin Dashboard

Manage users screen allows administrators to manage registered users, including suspending accounts, reviewing activities, and responding to reports.

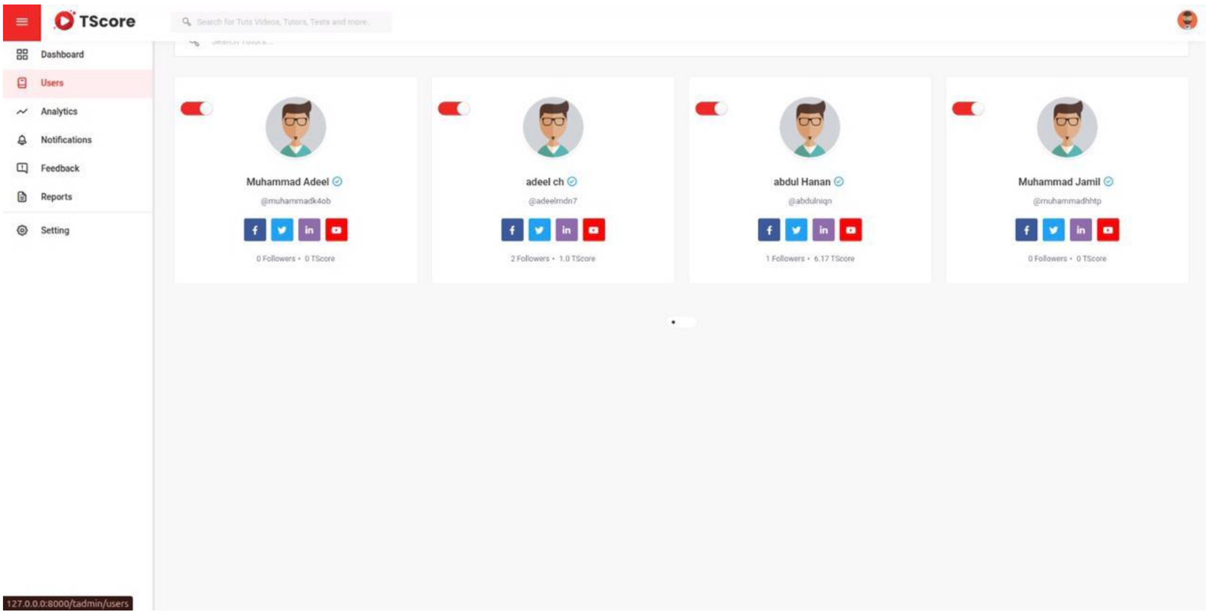

Figure 5.21: Manage Users By Admin

Screen shown in Figure 5.22 displays statistics and visualizations on user activity, most viewed videos, platform growth, and ad performance.

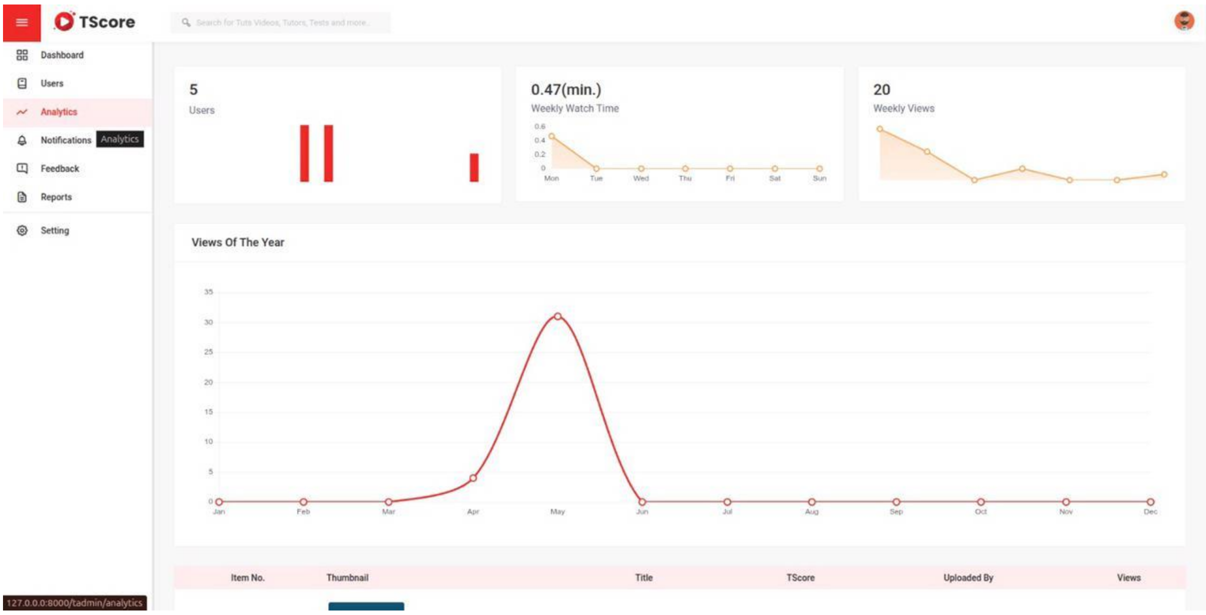

Figure 5.22: Website Analytics

Report action screen shows the actions taken by admin on reported content including warnings issued, content removals, and replies to reporters.

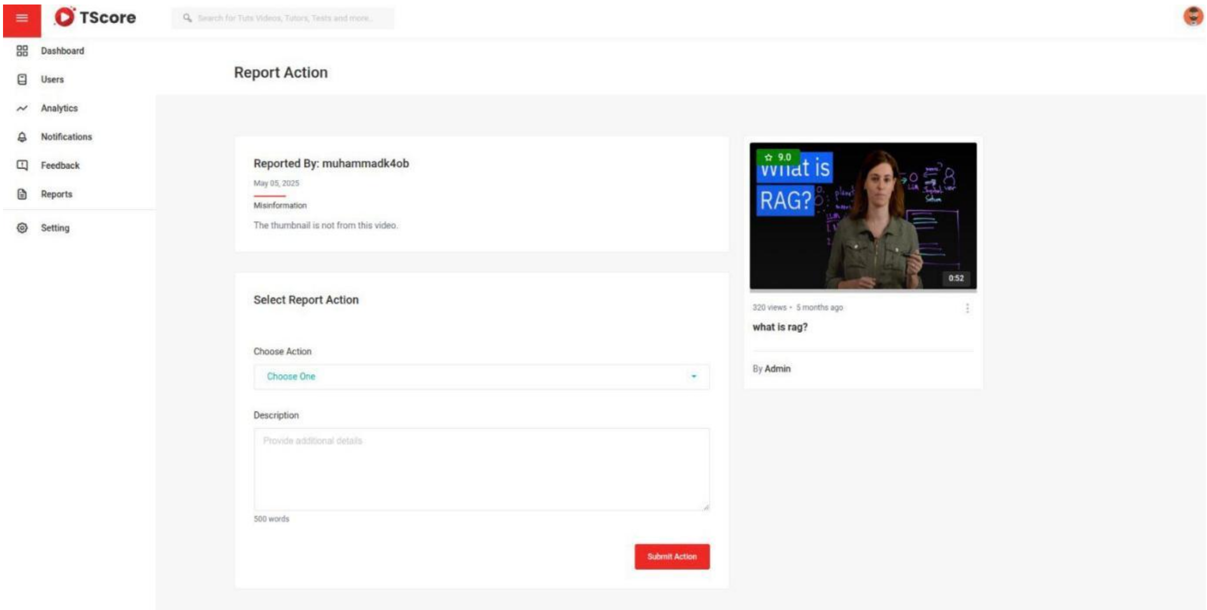

Figure 5.23: Report Action Screen

Notifications screen lists all alerts related to uploads, truth score updates, feedback replies, and account activities.

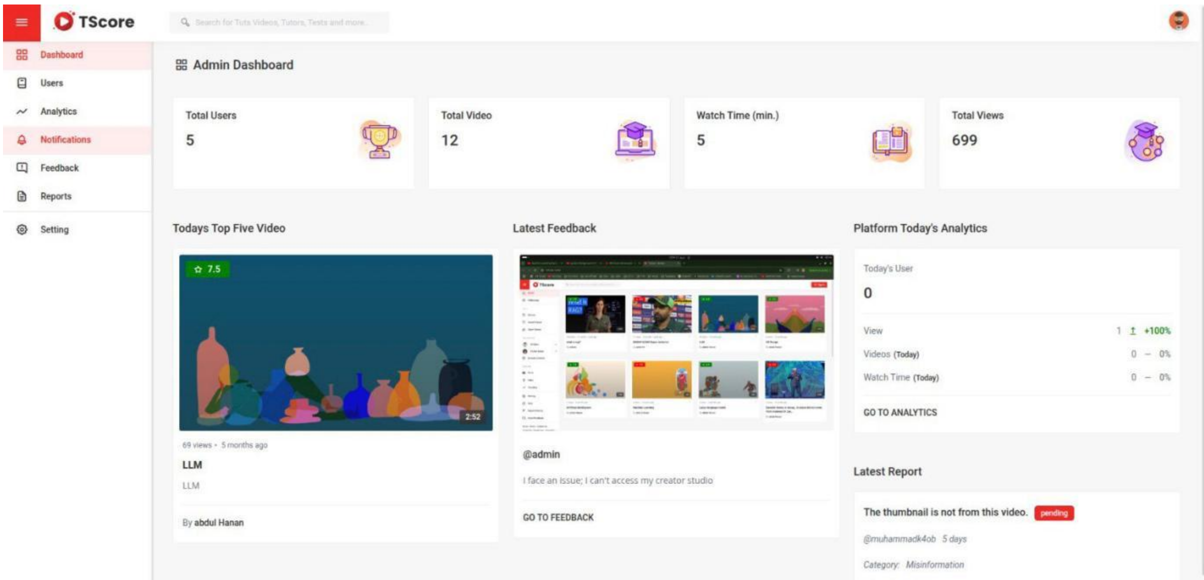

Figure 5.24: Admin Notifications

Reports by users provides a dashboard for admins to view and prioritize reports submitted by users about video content.

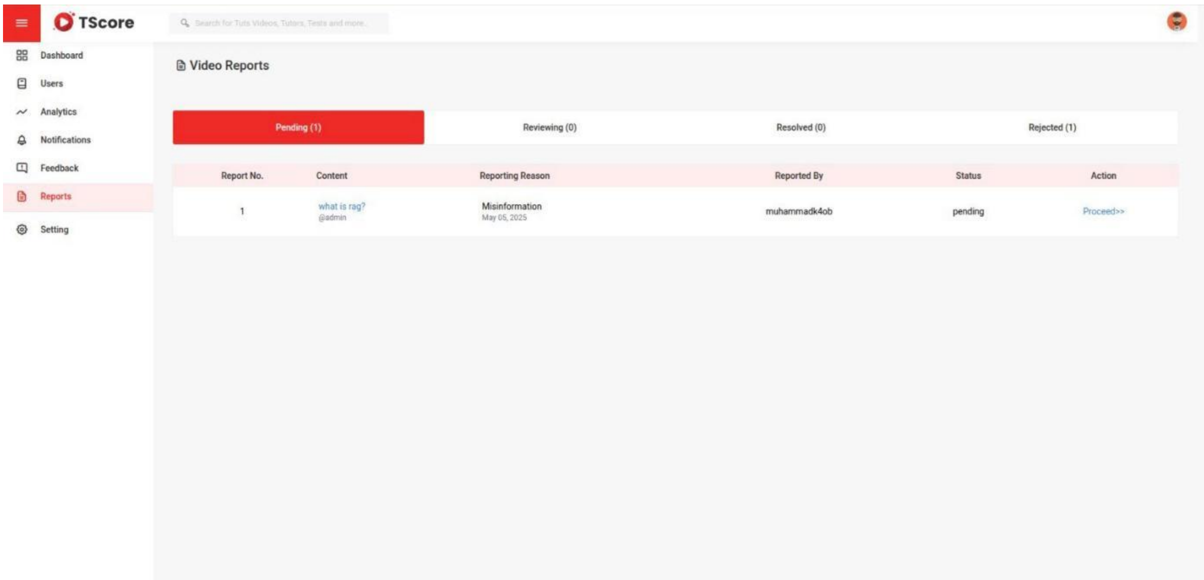

Figure 5.25: Reports By Users

Screen shown in Figure 5.26 displays feedback submitted by users, categorized into complaints, feature requests, and general suggestions.

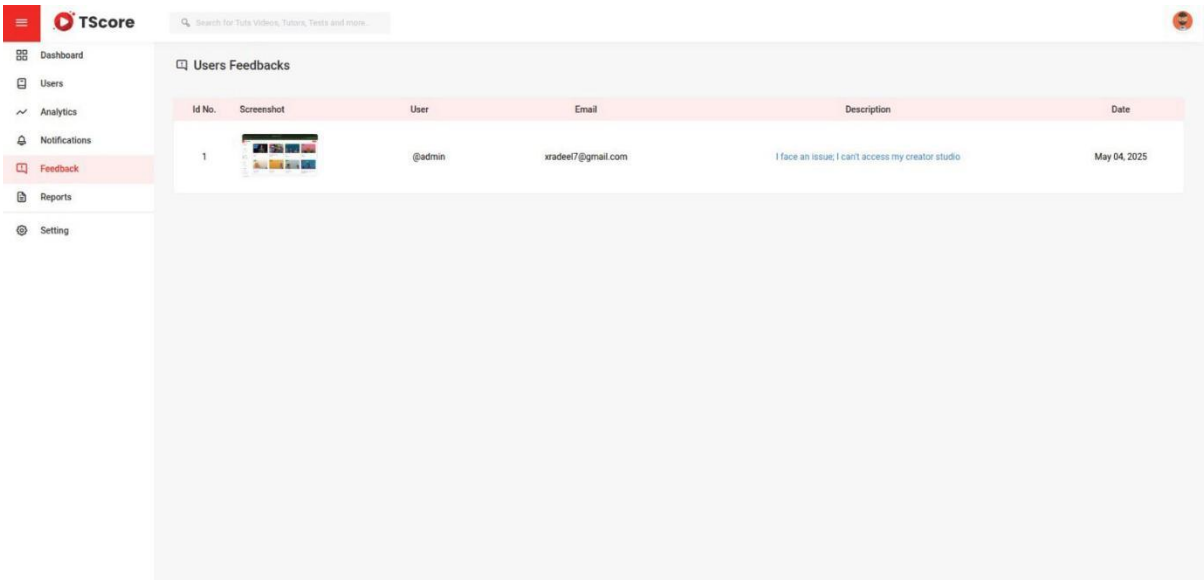

Figure 5.26: User Feedback

5.3.3 Creator's Dashboard

A dedicated panel for content creators showing their video performance, engagement metrics, and earnings.

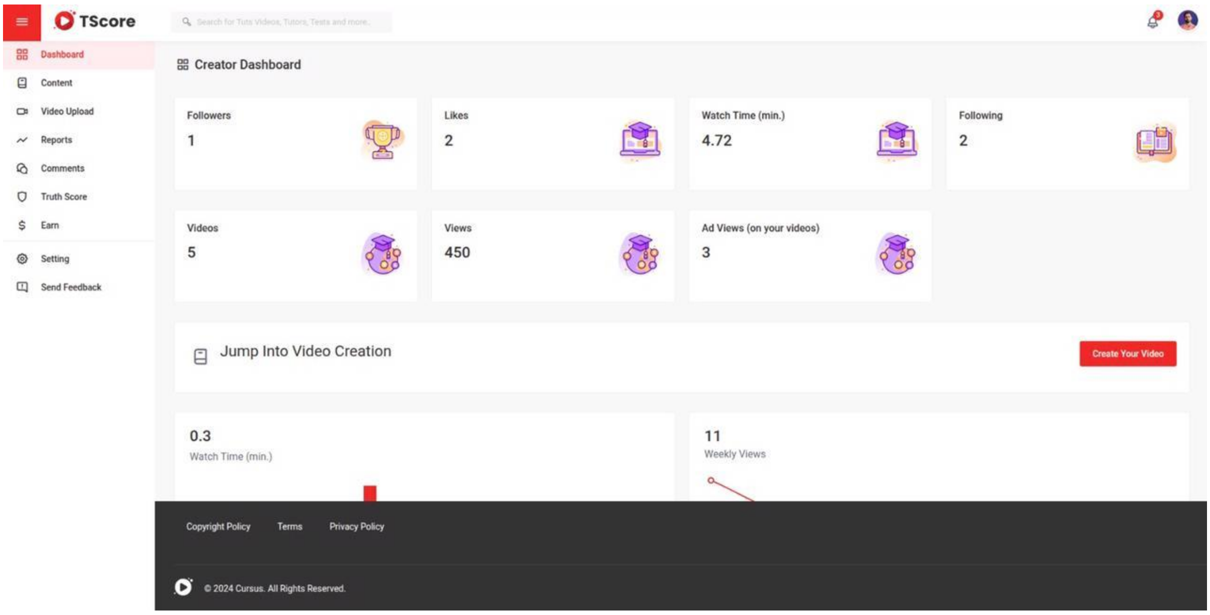

Figure 5.27: Creator's Dashboard

All comments are shown on this screen.

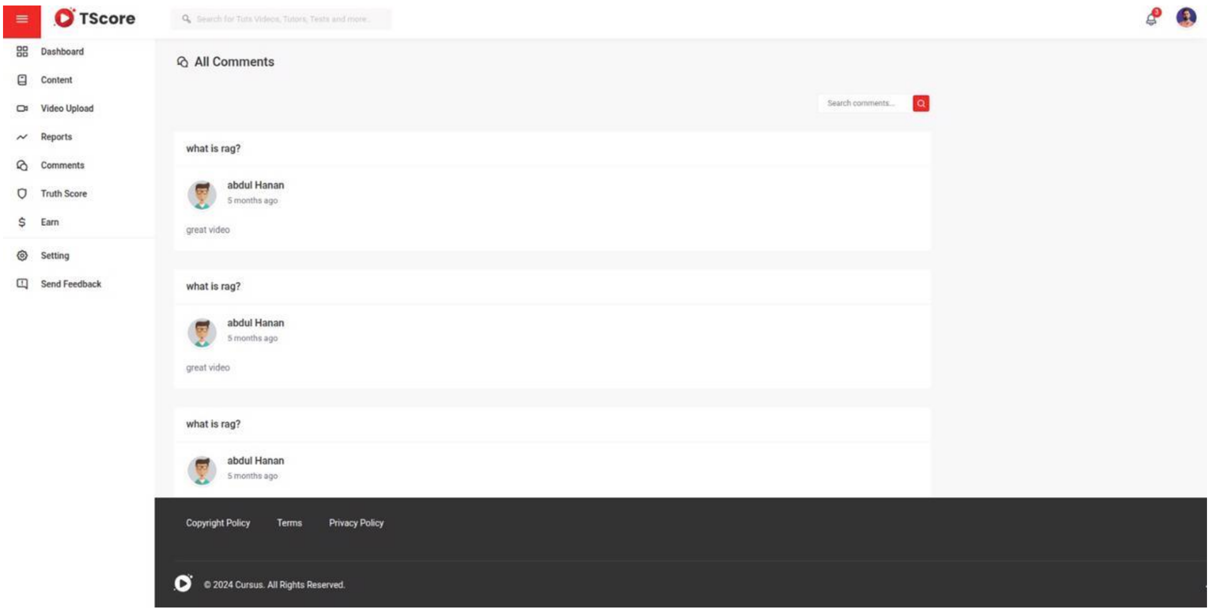

Figure 5.28: Comments Screen

Screen shown in Figure 5.29 lists all videos flagged by users for review, sorted by severity and frequency of reports.

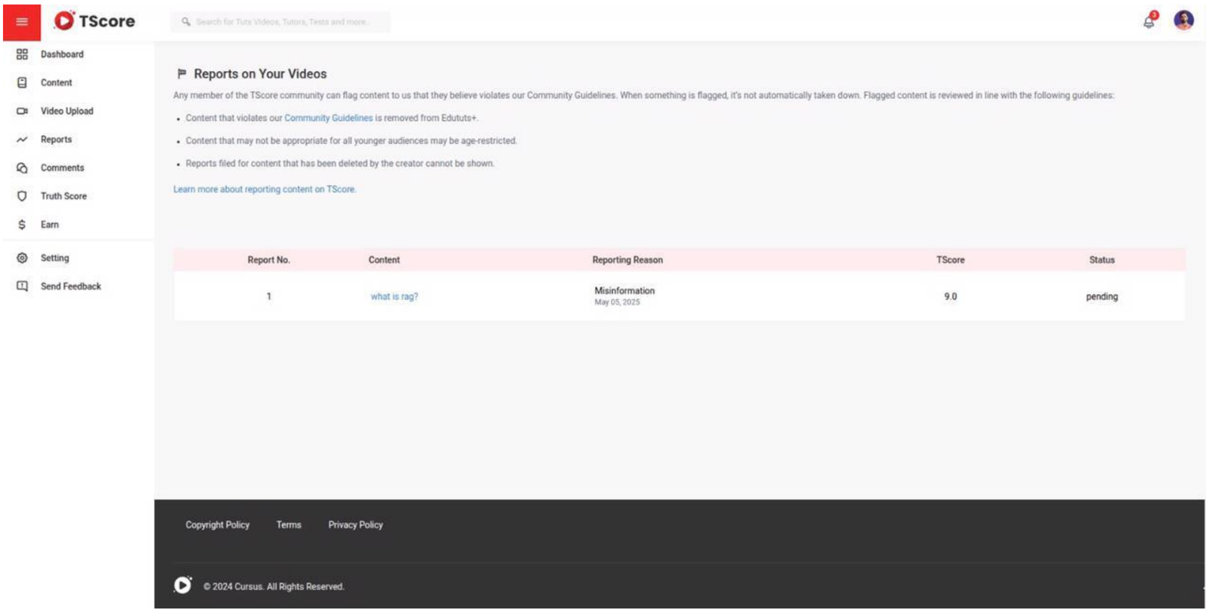

Figure 5.29: Reported Videos

Upload screen allows users to add new videos by providing a title, description, thumbnail selection, and selecting a file.

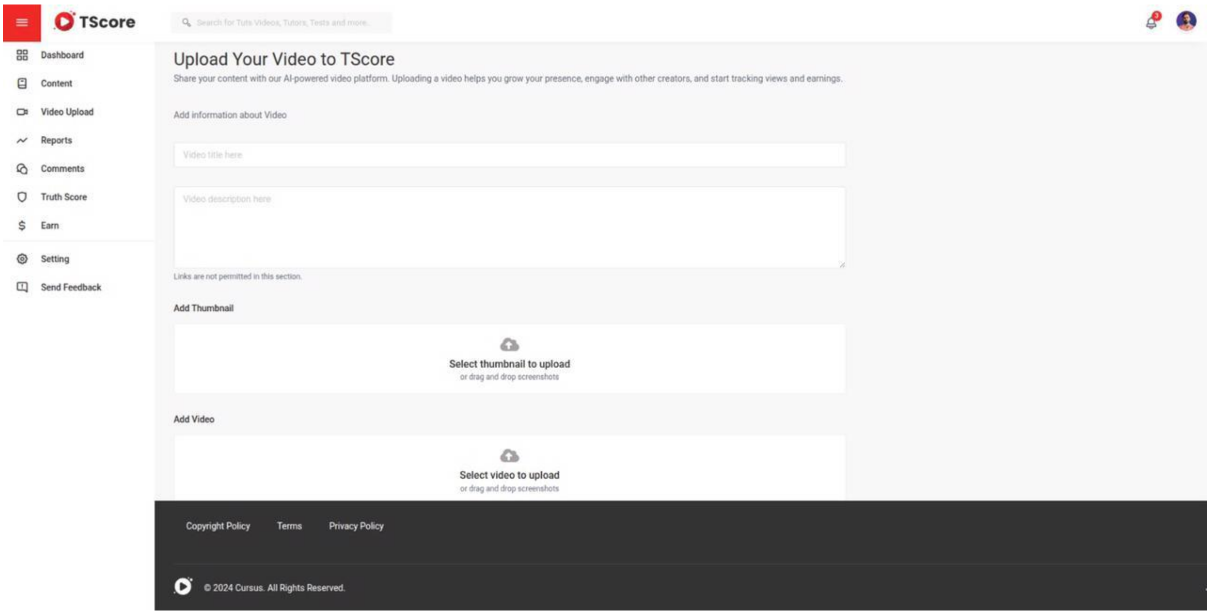

Figure 5.30: Upload New Video

Screen shown in Figure 5.31 displays the calculated truth score for each uploaded video with supporting details about how the score was derived.

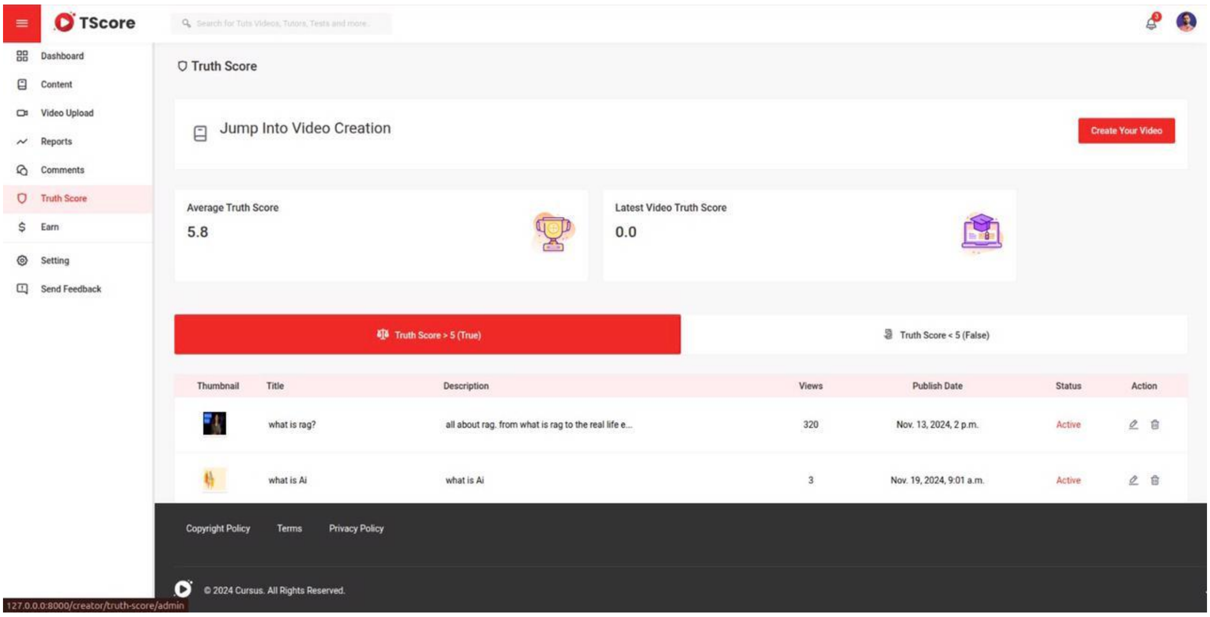

Figure 5.31: Truth Score Screen

Screen shown in Figure 5.32 shows a breakdown of a creator's earnings from ads, views, and bonuses based on content performance and ratings.

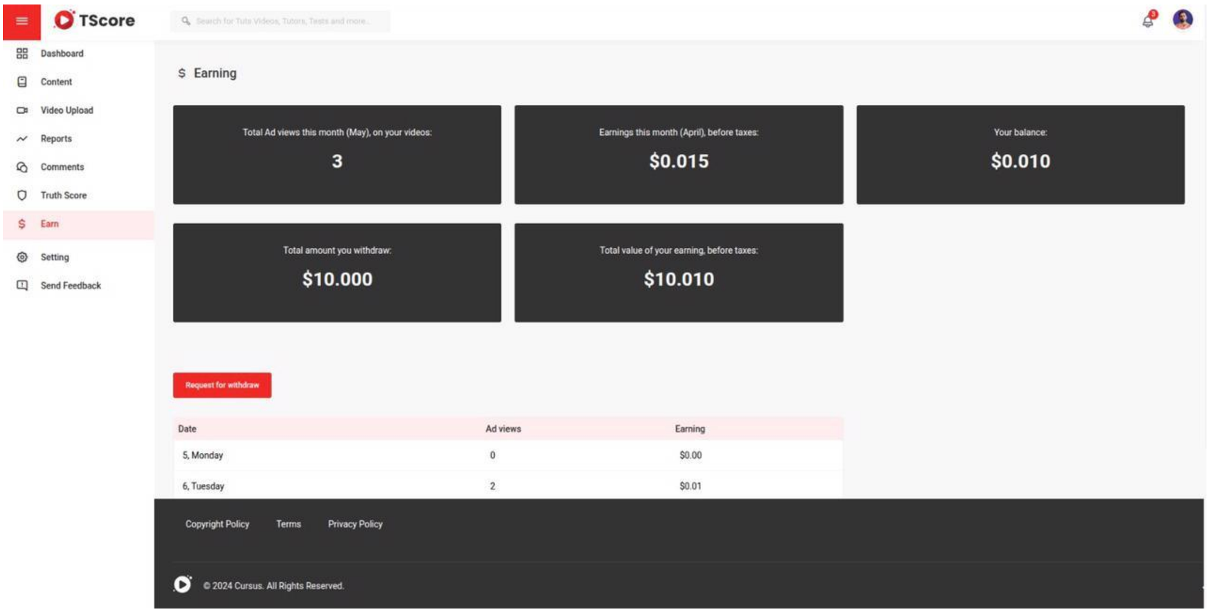

Figure 5.32: Creator's Earnings

Edit videos screen allows users or creators to edit previously uploaded videos, modify titles, descriptions, and update tags or change thumbnail.

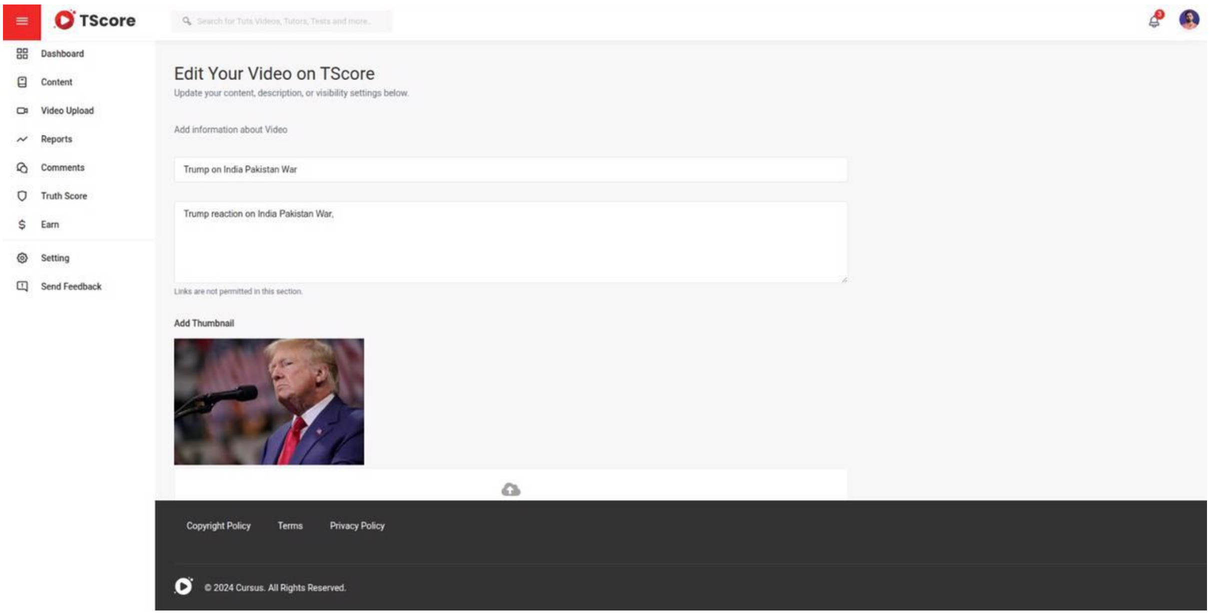

Figure 5.33: Edit Videos

Chapter 6

Testing and Evaluation

6 Testing

The testing and evaluation phase of TScore ensures that the developed system meets the required specifications, provides a smooth user experience, and performs all functionalities as expected. This chapter outlines various types of testing conducted during the project, including manual testing, unit testing, functional testing, integration testing, and automated testing.

6.1 Manual Testing

Manual testing involves verifying the system's behavior without using automated tools to ensure that each module of TScore behaves as intended.

Table 6.1: Manual Testing

Test Case	Expected Result	Actual Result	Status
User registration	User registered successfully	User registered successfully	Pass
User login	Successful login	Successful login	Pass
Video upload	Video is uploaded	Video is uploaded	Pass
Transcript generation	Transcript is auto-generated	Transcript generated successfully	Pass
Truth score evaluation	Truth score is displayed	Truth score is displayed	Pass
Select translation language	Drop-down with 10 top languages	Drop-down displayed correctly	Pass
User report videos	User report a video	User report a videosuccessfully	Pass
Admin login	Successful login	Successful login	Pass
Manage Users (Admin)	Users listed with actions (edit/delete)	Users managed successfully	Pass

Explaination

Manual testing for TScore was conducted to validate key functionalities of the platform by simulating user interactions and verifying if each feature behaves as expected. Below are the test cases that were manually executed to ensure core processes are working properly.

- **User Registration:** Verified that new users can register successfully using valid credentials. The system created user accounts and directed them to the login page after successful registration.
- **User Login:** Ensured that registered users can log in using their email and password. The system authenticated the credentials and redirected users to their dashboard.
- **Video Upload:** Tested the video upload functionality. Users were able to upload a video file along with metadata (title, description). The system stored the video and displayed it on the platform.
- **Transcript Generation:** After video upload, the system automatically generated a transcription using speech-to-text functionality. The accuracy and completeness of the transcription were validated.
- **Truth Score Evaluation:** The transcription was passed to Gemini for analysis. The system successfully retrieved and displayed a truth score along with a detailed explanation of the evaluation.
- **Language Selection for Translation**: Confirmed that users could select a language from a dropdown menu containing the top 10 most commonly used languages. The dropdown displayed correctly and allowed users to switch translation output.
- **Video Reporting by Users:** Users were able to report videos they found misleading or inappropriate. The system successfully recorded the report and notified the admin.
- **Admin Login:** Tested the admin login functionality to ensure only authorized users with admin roles could access the admin panel. Login was successful with valid admin credentials.
- **Manage Users:** Verified that the admin can view a list of all registered users and perform actions like editing or deleting user accounts. The system responded correctly and updated the user data as expected.

6.1.1 Unit Testing

Unit testing involves testing individual components or modules of the TScore system to ensure each part functions independently and correctly. Each unit is tested in isolation to validate its behavior and interaction with input data. The following tables represent unit tests for some core modules. [5]

Unit Testing 1: User Login

Testing Objective: To ensure that the login form is working correctly for users.

Table 6.2: Unit Testing 1

No.	Test Case/Test Script	Attribute and Value	Expected Result	Result
1	Verify user login with correct input	Username: user01, Password: abc123	User is successfully logged in and redirected to the dashboard	Pass
2	Attempt login with incorrect credentials	Username: user01, Password: wrong	Error message is displayed "Invalid credentials"	Pass

Unit Testing 2: Admin Login

Testing Objective: To ensure the admin login form is working properly.

Table 6.3: Unit Testing 2

No.	Test Case/Test Script	Attribute and Value	Expected Result	Result
1	Admin logs in with valid credentials	Username: admin, Password: admin123	Admin dashboard loads successfully	Pass
2	Admin tries to log in with invalid password	Username: admin, Password: wrongpass	Display error message	Pass

Unit Testing 3: Content Upload

Testing Objective: To ensure the video upload module accepts valid content and rejects invalid formats.

Table 6.4: Unit Testing 3

No.	Test Script	Attribute and Value	Expected Result	Result
1	Upload MP4 video	File: video.mp4	Content uploaded successfully	Pass
2	Upload unsupported format	File: document.pdf	Error message "Unsupported file format"	Pass

Unit Testing 4: Truth Score Evaluation

Testing Objective: To ensure the Gemini API and Langchain modules process and return truth scores correctly after the system generates the transcript.

Table 6.5: Unit Testing 4

No.	Test Case/Test Script	Attribute and Value	Expected Result	Result
1	Upload video and auto-generate transcript	Video File: AI_talk.mp4	Transcript generated and truth score returned with confidence level	Pass
2	Upload corrupted video file	Video File: corrupted_video.mp4	Error message "Unable to process video"	Pass

Unit Testing 5: Transcript Translation

Testing Objective: To ensure the translation module works correctly with supported languages

Table 6.6: Unit Testing 5

No.	Test Case/Test Script	Attribute and Value	Expected Result	Result
1	Translate English to Urdu	Input: Climate change affects weather	Translated Urdu transcript is shown	Pass

6.1.2 Functional Testing

Functional testing verifies whether TScore meets business and user requirements by ensuring that each feature works as intended. It involves testing core functionalities such as video uploading, truth score generation, multilingual transcription and translation, user authentication, and engagement features like commenting and sharing. This testing ensures proper input handling, accurate output, and correct system behavior under various conditions. The goal is to confirm that the system performs reliably, delivers a seamless experience, and aligns with platform objectives.

Table 6.7: Funtional Testing

Test Case ID	Test Case Name	Expected Results	Actual Result	Pass/Fail
FT-001	Home Page Access	Users are directed to the home page	Users are directed to the home page	Pass

Test Case ID	Test Case Name	Expected Results	Actual Result	Pass/Fail
FT-002	User Login	Users log in securely with correct credentials	Users log in securely	Pass
FT-003	Video Upload	Users upload video successfully	Video uploaded and stored	Pass
FT-004	Truth Score Generation	System analyzes and generates truth score	Score generated successfully	Pass
FT-005	Multilingual Translation	Transcripts translated into top 10 languages	Translations displayed correctly	Pass
FT-006	Comment and Engagement Feature	Users can like, comment, and share videos	All features function properly	Pass
FT-007	Profile Management	Users can update and manage their profile	Profile updated successfully	Pass
FT-008	Admin Video Moderation	Admins can delete/report violating videos	Videos moderated correctly	Pass
FT-009	Secure Logout	Users log out and session is terminated securely	Session ended successfully	Pass

Explaination

Functional testing for TScore was conducted to validate that the core features of the platform work as expected and align with user and business requirements. The testing began with Home Page Access (FT-001), ensuring that users were correctly directed to the home page upon accessing the platform. Next, User Login (FT-002) was tested to confirm that users could securely log in using valid credentials, with the system correctly authenticating users and granting access.

The Video Upload (FT-003) test case verified that users could upload videos along with titles and descriptions, and that the system successfully stored the media. Upon upload, the system automatically triggered the Truth Score Generation (FT-004) process. This test ensured that the transcription was processed and analyzed by the AI module (powered by Gemini), and a truth score was generated and displayed accurately.

To support global usability, Multilingual Translation (FT-005) was tested, verifying that transcripts were correctly translated into the top 10 widely used languages, with all translations appearing accurately. The Comment and Engagement Feature (FT-006) allowed users to interact with content by liking, commenting, and sharing videos, all of which were tested and confirmed to be functioning properly.

Profile Management (FT-007) was validated to ensure users could edit their profile information, such as username, bio, or profile picture, with updates saved successfully. On the administrative side, Admin Video Moderation (FT-008) was tested to confirm that admins could view, report, or delete inappropriate or violating content, and actions were properly reflected on the platform.

Finally, Secure Logout (FT-009) was tested to ensure that users could log out safely, with sessions terminated correctly to prevent unauthorized access. Overall, all functional test cases passed, indicating that TScore's primary user flows and features are working reliably and securely.

6.1.3 Integration Testing

Integration testing evaluates interactions between different TScore modules to ensure smooth data flow and system performance. [6]

Table 6.8: Integration Testing

Test Case ID	Test Case Name	Expected Result	Actual Result	Status
IT-001	Frontend-Backend Integration	Data flows seamlessly between UI and backend	API calls executed successfully	Pass
IT-002	Database Connectivity	Backend connects and performs CRUD on the database	Data saved and retrieved correctly	Pass
IT-003	Role-Based Access Control	Features restricted based on user roles (user/admin)	Access managed correctly	Pass
IT-004	Truth Score API Integration	NLP/ML models return score after video processing	Truth score generated accurately	Pass
IT-005	Translation Module Integration	Transcript passed to translation service and returned	Translations received	Pass

Explaination

Integration testing for TScore was carried out to validate the seamless interaction between the platform’s various components and ensure smooth data flow across the system. The first test case, Frontend-Backend Integration (IT-001), confirmed that the user interface and backend communicated effectively, with API calls functioning correctly and data being exchanged without disruption. Next, Database Connectivity (IT-002) tested the backend’s ability to perform Create, Read, Update, and Delete (CRUD) operations, ensuring that user data, video content, and other information were stored and retrieved accurately.

To validate security and permissions, Role-Based Access Control (IT-003) was tested. This ensured that different users, such as general users and administrators—had access only to features relevant to their roles. The Truth Score API Integration (IT-004) was a critical test validating that the AI models (leveraging Gemini) successfully processed the video transcription and returned a truth score, verifying the integration of NLP/ML services. Lastly, the Translation Module Integration (IT-005) confirmed that the transcription could be passed to the translation service and receive translations in the top 10 global languages accurately. Overall, all integration test cases passed, confirming that the different subsystems of TScore function together as a cohesive and reliable platform.

6.2 Automated Testing

Automated testing uses scripts and tools to test TScore modules regularly, ensuring consistent performance without manual intervention.

Table 6.9: Automated Testing

Tool Name	Tool Description	Applied on (Related Test Cases / FR / NFR)	Results
PyTest	Python testing framework used for unit and integration testing of backend modules	FR-001 (User Auth), FR-002 (Video Upload API), FR-008 (Truth Score Engine)	All unit & logic tests passed successfully
Selenium	Web automation tool for testing user flows in the frontend UI across browsers	FR-003 (Video Player UI), FR-006 (Search & Filter), FR-010 (Dashboard Navigation)	User flows executed without UI breakage on Chrome, Firefox
Postman	API testing tool used to validate HTTP endpoints, auth headers, and response data	FR-002 (Upload API), FR-004 (Translation API), FR-009 (Score Retrieval API)	API responses valid; latency within acceptable limits

Explaination

Automated testing was implemented for TScore to ensure system reliability, performance, and consistent user experience across platforms. The PyTest framework was used extensively for unit and integration testing of backend modules developed in Python. It was applied to key functional requirements such as user authentication (FR-001), video upload API (FR-002), and the truth score engine (FR-008). All logic-based tests passed successfully, confirming backend robustness. For frontend testing, Selenium was utilized to automate user interactions and verify UI consistency across different browsers. It validated features like the video player interface (FR-003), search and filter functionality (FR-006), and dashboard navigation (FR-010). These tests confirmed that user flows were executed smoothly without UI glitches in Chrome and Firefox. Additionally, Postman was employed for thorough API testing, focusing on endpoints such as the video upload (FR-002), translation (FR-004), and truth score retrieval (FR-009) APIs. The responses were accurate, and latency remained within acceptable performance limits. This layered automated testing approach ensured that TScore functions reliably from backend logic to user-facing features.

Chapter 7
Conclusion and Future Work

7 Conclusion and Future Work

7.1 Conclusion

The TScore system was developed to evaluate the reliability of spoken content in videos by generating a truth score using natural language processing and transcript-based analysis. This platform allows users to upload videos, after which the system automatically generates a transcript, translates it (if needed), and calculates a truthfulness score based on integrated verification mechanisms.

Throughout the development process, a strong emphasis was placed on usability, automation, and accurate content evaluation. The system successfully supports multiple user roles, handles backend transcript generation, and presents scores in a clear and intuitive interface. Manual and automated testing verified the functionality of all core features. The architecture was designed to allow scalability, security, and easy integration with external APIs or databases in future versions.

This project contributes toward combatting misinformation and empowering users to make better-informed decisions by offering a tech-driven solution for video truth verification. The overall implementation demonstrates the viability of automated fact analysis tools in content platforms.

7.2 Future Work

While the initial development of TScore has been successful, there is always room for improvement and innovation. Future work on the platform could focus on the following areas.

- **Appeal Mechanism:** Allow users to challenge or request a review of the assigned truth score with evidence or justification.
- **Extended Language Support:** Expand beyond the top 10 languages to support more regional and international audiences.
- **Mobile Application:** Build Android and iOS apps for greater accessibility and user engagement.
- **Content Categorization:** Automatically categorize videos by topic (e.g., politics, health, education) for better filtering.
- **Improved Truth Score Evaluation Engine:** Incorporate advanced AI models, to increase the accuracy and reliability of truth score calculations based on context, source credibility, and cross-referenced data.

By incorporating these future modules, TScore can evolve into a powerful, intelligent, and user-centric truth analysis platform trusted by a wider community. With enhanced AI capabilities, real-time fact-checking, multilingual support, and user-driven feedback systems, the platform can foster greater transparency, media literacy, and informed decision-making. These additions will not only improve user engagement and trust but also position TScore as a leading solution in the realm of AI-powered video credibility assessment. [7]

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